Milano-Bicocca University

Faculty of Mathematical, Physical and Natural Sciences

Degree in Material Science

SURFACE BASICITY MEASUREMENT OF REGULAR AND DEFECTIVE SITES OF BINARY OXIDES BY BF_3 ADSORPTION. QUANTUM SIMULATIONS.

Supervisor: Prof. Gianfranco Pacchioni

Co-supervisor: Dr. Cristiana Di Valentin

Thesis of:

Corrado Locati

N. 535774

Academic Year 2002–2003

This document is the English translation of my master's thesis in Material Science, originally defended at the University of Milano–Bicocca on 09/03/2004.

The thesis was translated personally by me. I extended Chapter 2, moreover on the contractions and the LYP model, with respect to the original text, for clarity. I fixed the bibliography, and I also removed the acknowledgement; I had fun writing them in a poem-fashion, something hard to reproduce in English, without giving any addition to the thesis work.

I assume full responsibility for any errors, omissions, or deviations from the original text. I apologize for any such occurrences.

Corrado Locati

22/07/2025

Table of Contents

1. Introduction

1.1 The Catalysis

In most industrial or biological processes, it is useful to accelerate the course of a reaction. The temperature can be increased, but it is an expensive and delicate operation as Many molecules can be decomposed.

Another pursuable path is the use of catalysts. A catalyst is a substance which increases the reaction speed without being consumed during the same. Every catalyst has a specific way of working, but all of them in general lower the activation energy of the process by opening a new reaction path with a different mechanism: a new lower–energy path. Another fundamental specification of a catalyst is his selectivity in the products: he must accelerate only the reaction that leads to desired product.

There are two types of catalysis: homogeneous, in which the catalyst is in the same phase as reagents and products, and heterogeneous, in which a reaction taking place in a separate phase is accelerated. A solid catalyst is usually used with liquid or gaseous reagents. The entire chemical industry would not exist as we know it today if there were no developments in heterogeneous catalysis.

Examples of reactions that use catalytic processes are:

- Decomposition of hydroiodic acid (HI) on gold.
- Decomposition of nitrogen dioxide (NO_2) on platinum.
- Addition of H_2 to double bonds $C = C$ (hydrogenation), to form simple bonds (useful in the oil industry, in the polymer and food industry, for the conversion of vegetable oil into margarine).

The reactions with heterogeneous catalysts take place at the interface between the two phases. The surface of a solid is a good site to deposit the catalyst, thus avoiding washing it or losing it among the obtained products.

Examples of reactions in which heterogeneous catalysis are:

- Haber-Bosch process:

$$N_2 + 3H_2 \rightarrow 2NH_3$$

with which low-cost fertilizers are produced. It uses iron and potassium adsorbed on a substrate of aluminum oxide (Al_2O_3).

- Fisher–Tropsch process

$$H_2 + CO + CH_3OH + fuel\ liquids$$

It uses iron and cobalt catalysts. This reaction is important because it provides fuel from biomass. It was used in autarchic regimes before and during the Second World War and more recently in South Africa.

- Three -way catalyst

$$NO_x, CO, HC \rightarrow H_2O + CO_2 + N_2$$

In this case, the right molecule is destroyed, instead of creating it. The exhaust gases of cars, mainly formed by carbon monoxide, non–stoichiometric nitrogen anhydrides and unburned hydrocarbons, pass through the three-way catalyst formed by a bed of platinum, rhodium, and palladium, to transform them into molecular nitrogen, water, and carbon dioxide.

1.2 Acid Sites and Basic Sites in Catalysis

Among the countless materials that can be used as catalysts there may be metals, semiconductors, or insulators. In the latter category oxides play a primary role in importance and study of their morphological properties has always been very alive and interesting. Their characterization, so the search for possible catalytic sites inside of surface defectiveness, is far from being finished. In this thesis work, the basicity of regular sites (terraces) and low coordination (LC) (edged, steps and vertices) on the surface of alkaline-earth oxides with face-centered cubic structure, MgO, CaO, SrO, BaO, and on TiO_2 and SiO_2 using BF_3 as a probe molecule.

Alkaline-earth ionic oxides of different nature have in common the fact of having centers of negative charge (the anions O^{2-}) which behave as the bases of Lewis, that is, capable of donating charge, and acid sites of Lewis (the cations M^{2+}), who receive charge. A greater or less acid or basic character is an indication of a greater or lesser goodness of that site in a catalytic process.

The problem of measuring the basicity and acidity of Lewis of oxides is well known and there are various methods and approaches that have been suggested in the past to achieve this result. The acidic strength of a solid is defined as the surface ability to convert a base electrically neutral in its conjugated acid, while with amount of acid (or "acidity"), the number of acid sites (or mmol) per unit of weight or per unit of surface area is meant.

To determine these quantities, both titrations with amines and indicators, and adsorption of gas-phase bases were used. These methods have the advantage of being also used at high temperature and during a catalytic process, but it is not possible to distinguish between a physical or chemical adsorption or differentiate between quantities of acid and various acid forces. To solve the last problem, it was decided to take advantage of the activity catalytic surfaces, but neither of the three methods is still able to distinguish between Brønsted and Lewis acids and this is useful for understanding the catalytic action of a surface acid[1].

Although not exclusive, the measure of Lewis acidity on the oxide surface is in general provided by CO and the consequent measure of the shift of the frequency of stretching of the $C-O$ bond induced by the bond with a surface cation[2]. CO is an excellent probe molecule as it shows a very small tendency to react, dissociate, polymerize, etc... Moreover, its vibrational frequency is very sensitive to small variations of the surrounding environment. A given surface cation, and the Infrared (IR) spectra provide a powerful tool to identify and classify the acidity of various sites. Very accurate analysis of the mechanisms that lie at the base of the shift in frequency ν of CO have been reported in the past and precise correlations have been established between the local electric field at the cation, its acidity of Lewis, and vibrational properties of the adsorbed CO.[3] Compared to this, the determination of Lewis's acidity using CO as a probe molecule can be considered a mature field.

Things are not so simple when you are interested in measuring surface basicity. Also in this case, basic force can be defined as the ability of a surface to convert an electrically neutral acid adsorbed into its conjugated base, that is, the ability of the surface of donating a pair of electrons to an adsorbed acid. The amount of basis of a solid is expressed as the number (or mmol) of basic sites per unit of weight or per unit of surface area. There are mainly two methods for the measurement of force and the quantity of basic sites: the title of benzoic acid using indicators e gaseous acid adsorption.

As for the first method, when a neutral acid indicator is adsorbed on a basic surface, the color of the indicator turns towards that of its married base. The basic force of the surface is therefore determined approximately observing the change of color in a $pK_a = pK_{BH}$ interval using benzene or iso–octane as solvents. The number of basic sites can instead be determined by titrating a suspension in benzene of a solid on which it was previously made to adsorb an indicator in its conjugated basic form, with benzoic acid dissolved in benzene. The benzoic acid titles are a measure of the number of basic sites that have a basic force corresponding to the pK_{BH} of the indicator used.[1]

The problems arise with the adsorption of acid in gaseous form. The first reason is that there is no simple binder that binds oxygen anions without reacting chemically to the surface, leading to strong geometric distortions of the absorbed and in fragmentation. A molecule that has been used for this purpose is CO_2. It reacts with the surface oxygen anions, O^{2-} to form surface carbonates CO_3^{2-}. The bond energy, the deviation from the linearity of the absorbed CO_2 molecule, the shift of the stretching mode of the C-O bond are all functions of the extent of the transfer of charge from the surface oxygen anion to the empty orbitals of the CO molecule[4,5].

Unfortunately, both the neutral CO_2 and the carbonate species CO_3^{2-} tend to bind to the surface oxygen in various ways, forming mono– and bidentate complexes and bringing therefore to significant complexity of the adsorbed species and of the resulting spectra[6]. While CO_2 was used to distinguish the reactivity and basicity of some simple oxide, it does not represent such a powerful and universal probe of the basicity of Lewis as the co is for the acidity of Lewis. Recently, the strong Lewis acid BF_3 was used to characterize the surface basicity of oxides such as Cr_2O_3 and SnO_2.[7,8]

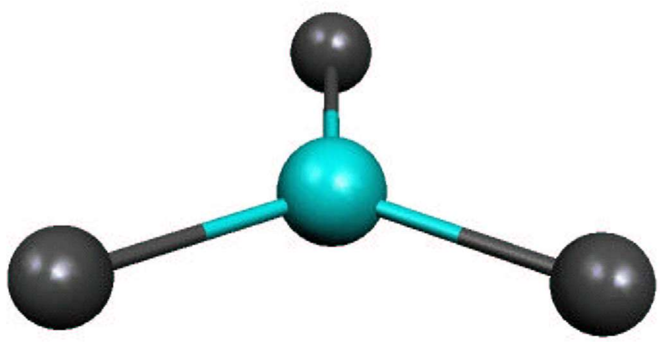

Figure 1.1: The BF_3 molecule; punctual group D_{3h}.

The BF_3 molecule was preferred to other molecules containing boron as it does not dimerize like BF_3 it does not polymerize like BI_3 and does not hydrolyze as BBr_3 and BCl_3. Thermal desorption spectra (TDS) and X–ray photo emission spectroscopy (XPS) were used to characterize the basicity of various sites on the two surfaces. In the past, BF_3 had already been used to measure the basicity of organic compounds[9,10]. BF_3 is also not an ideal probe molecule. In fact, at high temperatures it tends to dissociate itself on non–reactive surfaces, while it is fragmented on some oxide, even at low temperatures.[8] Despite these intrinsic problems, in principle BF_3 represents an interesting binder to monitor and to identify Lewis basicity sites on an oxide surface.

This work shows a direct correlation between the measurable properties of the adsorbed BF_3 (in particular, IR, XPS and TDS spectra) and indicators of the basicity of Lewis of a given site.

2. Computational methods

All calculations are based on the theory of the functional density (DFT) which in turn lies on the Born-Oppenheimer (B–O) approximation, which considers the big difference of mass between nuclei and electrons. The nuclei can be considered still and the Schrödinger equation for electrons in the static electric potential originating from the nuclei in given positions can be solved. Adopting various spatial nuclei displacements, the calculation is reiterated until building the surface of the potential energy of the system and identifying the equilibrium conformation as being the lowest point of this surface.

2.1 The Born- Oppenheimer Approximation

To better understand the functioning of the B–O approximation, we examine a unidimensional system such as the H_2^+ molecular ion, in which the motion is restricted to the direction of the X axis. The complete Hamiltonian $\widehat{H}$ of the problem is:

$$\widehat{H} = -\frac{\hbar^2}{2m_e}\frac{d^2}{dx^2} - \sum_{j=1}^{2}\frac{\hbar^2}{2m_j}\frac{d^2}{dX_j^2} + V(x, X_1, X_2) \tag{2.1}$$

Where:

- $-\frac{\hbar^2}{2m_e}\frac{d^2}{dx^2}$ is the kinetic energy of the electron.
- $-\sum_{j=1}^{2}\frac{\hbar^2}{2m_j}\frac{d^2}{dX_j^2}$ is the kinetic energy of the two nuclei.
- $+V(x, X_1, X_2)$ is the general potential energy.

Furthermore, x is the electron position and X_j with $j = 1, 2$ is the position of the two nuclei.

The Schrödinger equation is:

$$\widehat{H}\Psi(x, X_1, X_2) = \mathrm{E}\Psi(x, X_1, X_2) \tag{2.2}$$

A possible solution is:

$$\Psi(x, X_1, X_2) = \psi(x, X_1, X_2)\chi(X_1, X_2) \tag{2.3}$$

Where the full wavefunction $\Psi(x, X_1, X_2)$ is factorized in the electronic part, $\psi(x, X_1, X_2)$, which depends parametrically on nuclear coordinates, and nuclear part, $\chi(X_1, X_2)$, vibrational, rotational motion of the nuclei. With $\psi(x, X_1, X_2)$ it is intended that the electronic wave function depends in parametric form from the coordinates of the two nuclei X_1, X_2. That is, a different wave function $\psi(x)$ is obtained for each displacement of the two nuclei.

By substituting Equation 2.3 in Equation 2.2, and since the Hamiltonian is given by Equation 2.1, we obtain

$$\widehat{H}\psi\chi = -\frac{\hbar^2}{2m_e}\frac{d^2\psi}{dx^2}\chi - \sum_{j=1}^{2}\frac{h^2}{2m_j}\frac{d^2\chi}{dX_j^2}\psi - \sum_{j=1}^{2}\frac{h^2}{2m_j}\frac{\partial^2\psi}{\partial X_j^2}\chi - \sum_{j=1}^{2}\frac{h^2}{2m_j}\left(2\ \frac{\partial\psi\partial\chi}{\partial X_j^2}\right) + V(x, X_1, X_2)\psi\chi$$

$$\widehat{H}\psi\chi = -\frac{\hbar^2}{2m_e}\frac{d^2\psi}{dx^2}\chi - \sum_{j=1}^{2}\frac{h^2}{2m_j}\frac{d^2\chi}{dX_j^2}\psi - \sum_{j=1}^{2}\frac{h^2}{2m_j}\left(2\ \frac{\partial\psi\partial\chi}{\partial X_j^2} + \frac{\partial^2\psi}{\partial X_j^2}\chi\right) + V(x, X_1, X_2)\psi\chi \qquad 2.4$$

Where:

- $-\frac{\hbar^2}{2m_e}\frac{d^2\psi}{dx^2}\chi$ is the kinetic energy of the electron with coordinate x, acting only on ψ.
- $-\sum_{j=1}^{2}\frac{h^2}{2m_j}\frac{d^2\chi}{dX_j^2}\psi$ is the kinetic energy of the nuclei, acting only on χ.
- $-\sum_{j=1}^{2}\frac{h^2}{2m_j}\frac{\partial^2\psi}{\partial X_j^2}\chi$ is the contribution from the nuclear kinetic energy operator acting on the electron wavefunction ψ. This arises when ψ depends parametrically or explicitly on X_j.
- $-\sum_{j=1}^{2}\frac{h^2}{2m_j}\left(2\ \frac{\partial\psi\partial\chi}{\partial X_j^2}\right)$ is the cross-derivative terms arising by mixing the nuclear and electronic wave functions due to shared dependence on X_j.
- $V(x, X_1, X_2)\psi\chi$ is the potential energy term. It could include electron–nucleus attraction, nucleus–nucleus repulsion, etc., acting multiplicatively on the wavefunction.

More compactly:

$$\widehat{H}\psi\chi = -\epsilon'\psi\chi - \sum_{j=1}^{2}\frac{h^2}{2m_j}\left(2\ \frac{\partial\psi\partial\chi}{\partial X_j^2} + \frac{\partial^2\psi}{\partial X_j^2}\chi\right) \qquad 2.5$$

Where ϵ' it is the eigenvalue of the equation

$$-\sum_{j=1}^{2}\frac{h^2}{2m_j}\frac{d^2\chi}{dX_j^2} + \mathrm{E}(\mathrm{X}_1, \mathrm{X}_2)\chi = \epsilon'\chi \qquad 2.6$$

And E it is the eigenvalue that is obtained by solving:

$$-\frac{\hbar^2}{2m_e}\frac{d^2\psi}{dx^2}+V(x,X_1,X_2)\psi=\mathrm{E}(X_1,X_2)\psi \tag{2.7}$$

The approximation of Born-Oppenheimer consists in placing qual to zero the following term.

$$-\sum_{j=1}^{2}\frac{\hbar^2}{2m_j}\left(2\,\frac{\partial\psi\partial\chi}{\partial X_j^2}+\frac{\partial^2\psi}{\partial X_j^2}\chi\right) \tag{2.8}$$

which appears in the Schrödinger Equation 2.5. In reality, it is not exactly zero, but it is very small compared to $\epsilon'\psi\chi$ as the nuclear masses appear in the denominator, so it can be neglected. In this way, Equations 2.6 and 2.7 become solutions to the problem.

Equation 2.6 represents the Schrödinger equation for the wavefunction χ of the nuclei when the potential energy has the shape of the potential energy of the system. Its eigenvalue is the total energy of the molecule within the limits of the Born-Oppenheimer approximation.

Equation 2.7 is the Schrödinger equation related to an electron placed in the potential $V(x,X_1,X_2)$ which depends on the fixed position of the two nuclei. Its solution is the electronic wavefunction ψ, and the eigenvalue $\mathrm{E}(X_1,X_2)$ is the electronic contribution to the total energy, plus the potential energy of the internuclear repulsion. It is precisely this function which provides the curve of potential energy whose minimum corresponds to the equilibrium.

More generally, the Hamiltonian for a set of nuclei positions $\boldsymbol{r}$ is:

$$\hat{H}=-\frac{\hbar^2}{2m_e}\sum_{i=1}^{n}\nabla_i^2-\sum_{i=1}^{n}\sum_{I=1}^{N}\frac{Z_Ie^2}{4\pi\epsilon_0r_{Ii}}+\frac{1}{2}\sum_{\substack{i,j=1,\\ i\neq j}}^{n}\frac{e^2}{4\pi\epsilon_0r_{ij}} \tag{2.9}$$

Where:

- $-\frac{\hbar^2}{2m_e}\sum_{i=1}^{n}\nabla_i^2$ is the Kinetic energy of the electrons, with ∇_i^2 being the Laplacian acting on electron i.
- $-\sum_{i=1}^{n}\sum_{I=1}^{N}\frac{Z_Ie^2}{4\pi\epsilon_0r_{Ii}}$ is the electron–nucleus Coulomb attraction term, where Z_I is the charge of the nucleus I and r_{Ij} the distance between electron i and nucleus I.
- $+\frac{1}{2}\sum_{\substack{i,j=1,\\ i\neq j}}^{n}\frac{e^2}{4\pi\epsilon_0r_{ij}}$ is the electron–electron repulsion term, with ½ to avoid double-counting.

Usually in the Hamiltonian $\hat{H}$ the energy of internuclear repulsion is excluded and it is added afterwards as a classic term, at the end of the calculation. The solution to this equation is obtained with two methods:

- *Ab-initio* method, in which a model for the electronic wave function is chosen, and Equation 2.9 is solved, using only the values of the fundamental constants and the atomic number of the nuclei. The accuracy of this method is mainly determined by model chosen for the wave function.
- Semi-empirical methods, which exploit a simplified form of the Hamiltonian and adjustable parameters whose value has been obtained experimentally.

2.2 The Hartree-Fock Method

An *ab-initio* approach to the problem is represented by the Hartree-Fock (HF) method, in which each electron is considered moving in the field of nuclei and in the average field of the other n - 1 electrons. We are looking for a solution in the form of a Slater determinant:

$$\psi(1,2,\dots,N) = \left(\frac{1}{N!}\right)^{\frac{1}{2}} \begin{vmatrix} \phi_a(1) & \phi_b(1) & \cdots & \phi_z(1) \\ \phi_a(2) & \phi_b(2) & \cdots & \phi_z(2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_a(N) & \phi_b(N) & \cdots & \phi_z(N) \end{vmatrix} \quad 2.10$$

Where $\phi_i(n)$ is the nth spin orbital and the index i includes both the spin and the spatial states. The spin orbitals are calculated with the variational theory that involves the Search for Rayleigh's minimum ratio:

$$\epsilon = \frac{\int \psi^* \hat{H} \psi dx}{\int \psi^* \psi dx} \quad 2.11$$

By letting the wavefunctions ψ be dependent by one or more parameters to be minimized under the orthonormality constraint of the spin orbitals. From the application of the procedure, the Hartree-Fock for individual spin orbitals arise:

$$f_i \phi_i(n) = \epsilon_i \phi_i(n) \quad 2.12$$

With ϵ_i the orbital energy of the spin orbital $\phi_i(n)$ and f_i the Fock operator who considers both the electrostatic repulsion among the electrons and the exchange due to the spin correlation. A single spin orbital is obtained by solving the Equation 2.12 with the correspondent Fock operator

f_i. The latter, however, depends on the spin orbital of the other n - 1 electrons; it is as if to look for a solution, all the other solutions are needed. In practice, the process starts with a test ensemble of spin orbitals with which the Fock operator is formed, then the HF equations are solved by obtaining a set of spin orbitals with which it is built a better Fock operator than the previous one and so on. The process is completed when the solutions are auto coherent, that is, when the convergence criterion is satisfied.

2.3 The Density functional Theory

The biggest limit of the HF method lies in the fact that it does not consider the electronic correlation. Thanks to the two theorems of Hohenberg and Kohn of 1964, and to the computational method of Kohn and Sham (KS) of the following year, it is shown that the energy of an electronic system can be formulated according to the electronic probability density ρ. In an *n*-electrons system, $\rho(\boldsymbol{r})$ is the total electronic density in point $\boldsymbol{r}$. Electronic energy is defined as *functional of electronic density* ($E[\rho]$), in the sense that at a given function of the density $\rho(\boldsymbol{r})$ corresponds only one energy.

The exact energy of the fundamental state of a n-electron system in the Kohn–Sham density functional theory (DFT) can be written as:

$$E[\rho] = -\frac{\hbar^2}{2m_e}\sum_{i=1}^{n}\int \psi_i^*(\boldsymbol{r_1})\nabla_1^2\psi_i(\boldsymbol{r_1})\,\mathrm{d}\boldsymbol{r_1} - \sum_{I=1}^{n}\int \frac{Z_I e^2}{4\pi\epsilon_0 r_{I1}}\rho(\boldsymbol{r_1})\mathrm{d}\boldsymbol{r_1} + \frac{1}{2}\int\int \frac{\rho(\boldsymbol{r_1})\rho(\boldsymbol{r_2})\mathrm{e}^2}{4\pi\epsilon_0 r_{12}}\mathrm{d}\boldsymbol{r_1}\mathrm{d}\boldsymbol{r_2} + E_{XC}[\rho] \qquad 2.13$$

Where the monoelectronic Kohn–Sham spatial orbitals ψ_i $(\mathrm{i} = 1, 2, \ldots, \mathrm{n})$ are the solutions of the Kohn-Sham equations (later reported) for the electron i. $\rho(\boldsymbol{r})$ is the electron density, Z_I is the nuclear charge of nucleus I, r_{I1} is the distance between electron at r_1 and nucleus I; r_{12} is the distance between electrons at r_1 and r_2.

Furthermore, in Equation 2.13,

- $-\frac{\hbar^2}{2m_e}\sum_{i=1}^{n}\int \psi_i^*(\boldsymbol{r_1})\nabla_1^2\psi_i(\boldsymbol{r_1})\,\mathrm{d}\boldsymbol{r_1}$ is the kinetic energy of the electrons (as Kohn–Sham orbitals).
- $-\sum_{I=1}^{n}\int \frac{Z_I e^2}{4\pi\epsilon_0 r_{I1}}\rho(\boldsymbol{r_1})\mathrm{d}\boldsymbol{r_1}$ is the electron–nucleus attraction for all n nuclei.

- $+\frac{1}{2}\int\int \frac{\rho(\boldsymbol{r_1})\rho(\boldsymbol{r_2})\mathrm{e}^2}{4\pi\epsilon_0 r_{12}}\mathrm{d}\boldsymbol{r_1}\mathrm{d}\boldsymbol{r_2}$ is the Hartree electron–electron electrostatic repulsion between the total charge distribution in $\boldsymbol{r_1}$ and $\boldsymbol{r_2}$.
- $+E_{XC}[\rho]$ is the (approximate) exchange–correlation functional.

The exchange–correlation functional is also a function of the electronic density and considers all the non–classical interactions between electrons. Unfortunately, the Hohenberg–Kohn theorem of tells that E and therefore E_{XC} must be functional of the electronic density, but it does not refer to what analytical form they must have, so we are forced to adopt approximate forms.

The exact charge density of the fundamental state is given by:

$$\rho(\boldsymbol{r}) = \sum_{i=1}^{n}|\psi_i(\boldsymbol{r})|^2 \qquad 2.14$$

where the sum extends to all the occupied KS orbitals.

As previously mentioned, the KS orbitals are solutions of the KS equation, the effective one-electron Hamiltonian:

$$\left\{-\frac{\hbar^2}{2m_e}\nabla_1^2 - \sum_{I=1}^{n}\frac{Z_I e^2}{4\pi\epsilon_0 r_{I1}} + \int\frac{\rho(\boldsymbol{r_2})e^2}{4\pi\epsilon_0 r_{12}}d\boldsymbol{r_2} + V_{XC}(\boldsymbol{r_1})\right\}\psi_i(\boldsymbol{r_1}) = \epsilon_i\psi_i(\boldsymbol{r_1}) \qquad 2.15$$

Where:

- $-\frac{\hbar^2}{2m_e}\nabla_1^2$ is the kinetic energy operator for electron i.
- $-\sum_{I=1}^{n}\frac{Z_I e^2}{4\pi\epsilon_0 r_{I1}}$ is the electron nucleus attraction contribution.
- $\int\frac{\rho(\boldsymbol{r_2})e^2}{4\pi\epsilon_0 r_{12}}d\boldsymbol{r_2}$ is the Hartree potential, the electron-electron repulsion, the classical electrostatic field of all electrons.
- $V_{XC}(\boldsymbol{r_1})$ is the exchange–correlation potential, the quantum many–body effects.
- $\psi_i(\boldsymbol{r_1})$ is the i-th KS orbital.
- ϵ_i is the corresponding i-th KS orbital energy eigenvalue.

The exchange and correlation potential V_{XC} is the functional derivative of the exchange and correlation energy:

$$V_{XC}[\rho] = \frac{\delta E_{XC}[\rho]}{\delta\rho} \qquad 2.16$$

If E_{XC} is known, V_{XC} is also obtained. KS equations are resolved in an auto consistent fashion, as in the case of HF equations. First an approximated form of the E_{XC} is assumed, then V_{XC} as function of $\boldsymbol{r}$ is calculated. The system of equations of KS is then resolved to obtain an initial set of KS orbitals. From this ensemble we start again to calculate a more acceptable density with Equation 2.14, and the process is repeated until density and energy correlation energy converge within a given tolerance. Finally, from Equation 2.13, electronic energy is obtained.

2.3.1 LDA and LSDA methods

The main problem of the functional density theory is the search for a valid expression of the exchange and correlation energy E_{XC} of the system. Several systems were investigated to obtain the approximate forms of E_{XC}.

In the local density approximation (LDA),

$$E_{XC} = \int \rho(\boldsymbol{r})\epsilon_{XC}|\rho(\boldsymbol{r})|\,\mathrm{d}\mathbf{r} \qquad 2.17$$

Where $\epsilon_{XC}|\rho(\boldsymbol{r})|$ is the exchange and correlation energy in a homogeneous gas of electrons of constant density (Jellium).

In the local spin density approximation (LSDA), the electronic density is divided into two terms, one for each spin configuration. In closed–shell systems, a LSDA gives the same results as the LDA. In a hypothetical Jellium, an infinite number of electrons travel through a uniform and continuous positive charge distribution, built to preserve electroneutrality. If LDA is applied, it is assumed that in every point of the space the exchange and correlation functional has the same form as Jellium with the same electronic density, or that the exchange and correlation functional varies very slowly. However, in real systems, the positive charge is not evenly distributed as in the Jellium model, but it is located only around the nuclei, therefore the density varies sharply and LDA calculations are often not in good agreement with the experimental results.

More accurate methods must overcome the condition of non–uniformity of the electron gas.

2.3.2 Generalized Gradient Approximation (GGA)

The first step to overcome the approximation to Jellium density is to consider the exchange and correlation energy not only as function of the electronic density, but also of its derivatives with respect to the nuclear coordinates. This leads to the Generalized Gradient Approximation (GGA), which better accounts for density inhomogeneities present in real molecular and solid-state systems.

Among the various functional built with the GGA, the Becke functional which provides a gradient correction to the LSDA exchange energy[11] and the Lee–Yang–Parr which offers an accurate formulation of the correlation energy.[12]

2.3.3 B3LYP Method

A further refinement to solve the problem of searching for a suitable E_{XC} expression is represented by hybrid methods. In these methods, the exchange term is partly calculated with the HF method, and partly with the DFT method. An example is the B3LYP method, in which the exchange and correlation term assumes the form:

$$E_{XC}^{B3LYP} = AE_X^{X\alpha} + (1-A)E_X^{HF} + B\Delta E_X^{Beche} + E_C^{VWN} + C\Delta E_c^{LYP} \quad 2.18$$

Where:

- $E_X^{X\alpha}$or E_X^{LDA} is the LDA exchange energy (Slater or Dirac functional) with coefficient $\alpha = 2/3$.
- E_X^{HF} is the exact Hartree-Fock exchange energy.
- ΔE_X^{Beche} is the difference between the three-parameters Becke exchange functional (B3)[12] and the exact term HF, also named gradient correction to exchange from Becke 1988.
- E_C^{VWN} is the correlation energy from Vosko–Wilk –Nusair LDA functional.
- ΔE_C^{LYP} It is the non–local electronic correlation part calculated according to the Lee, Yang, Parr (LYP) expression[13], also named as gradient correction to correlation from Lee–Yang–Parr.
- A, B, C are empirical parameters fitted to reproduce experimental thermochemical data.

The values of the three parameters A, B and C are those calculated by Becke through a fitting of energy and potential ionization for different atomic systems.

- A (exact exchange weight) = 0.20.
- B (Becke exchange correlation) = 0.72
- C (LYP correlation) = 0.81.

In this thesis, the calculations are based on the theory of the functional density and the functional of the correct exchange gradient with the three Becke parameters in combination with the functional correlation of Lee, Yang and Parr (B3LYP).

The B3LYP functional mixes:

- A portion of Hartree–Fock exchange $E_X^{HF}$with DFT exchange (LDA $E_X^{X\alpha}$ + Becke ΔE_X^{Beche}),
- Combines LDA correlation (VWN E_C^{VWN}) with a gradient-corrected LYP correlation ΔE_C^{LYP}.

2.4 Basis Functions

So far, we have described various methods with which to find the potential energy surface of the system. To do this, it is necessary to describe the orbitals with special sets of basis functions. Molecular orbitals are commonly expressed as linear combinations of atomic orbitals (LCAO):

$$\psi_i(\boldsymbol{r}) = \sum_{\mu=1}^{n} c_{\mu i}\, \phi_\mu(\boldsymbol{r}) \qquad 2.19$$

Where:

- $\psi_i(\boldsymbol{r})$ is the *i*-th molecular orbital.
- $\phi_\mu(\boldsymbol{r})$ are the basis functions.
- $c_{\mu i}$ are the molecular orbital coefficients to be determined.

Atomic orbitals (AO) are solutions of the equations of monoelectronic Hartree-Fock. The atomic functions $\phi_\mu(\boldsymbol{r})$ were in the beginning described through Slater–type orbitals (STO) for their close resemblance to the hydrogen-like atomic orbitals. STOs are defined in spherical coordinates:

$$\phi_i(\zeta; n, l, m; r, \theta, \varphi) = N r^{n-1} e^{-\zeta r} Y_{lm}(\theta, \varphi) \qquad 2.20$$

Where:

- N is the normalization constant.
- ζ is the orbital exponent, which controls radial decay.
- n, l, m are the principal, angular momentum, magnetic momentum quantum numbers, respectively.
- r, θ, φ are the spherical coordinates.
- $Y_{lm}(\theta, \varphi)$ are the spherical harmonics (angular part of the wavefunction).

Unfortunately, the use of the STA is limited as they are not suitable for calculations of bi–electronic integrals, therefore the Gaussian type orbitals, GTOs were introduced.[14] STOs can be approximated by a linear combination of GTOs, each with different exponents and coefficients. A primitive cartesian GTO has the following form:

$$g(\alpha; l, m, n; x, y, z) = N e^{-\alpha r^2} x^l y^m z^n \qquad 2.21$$

Where:

- N is the normalization factor.
- α is the orbital exponent, which controls the width of the Gaussian.
- l, m, n are the powers of the cartesian coordinates. They define the angular momentum.
- $r^2 = x^2 + y^2 + z^2$ is the square of the radial distance from the origin.

The simplest basis set is the minimal one (MB), in which only one function is used to represent each of the orbitals. Unfortunately, with such a poor basis set, it is not possible to make accurate calculations. To reach energy values close to the Hartree–Fock limit (given by the variational expression 2.11 which does not consider the electronic correlation), wider basic sets are then needed. In molecular calculations, GTOs must be contracted[15], which means that appropriate linear combinations of GTOs are used as basic functions.

2.4.1 Contractions and Their Notation

The number of contractions used to represent a single STO (i.e., ζ) is an indication of the goodness of the set. There may be single zeta basis (SZ), double zeta (DZ), triple zeta (TZ), and so on. Increasing the zeta level enhances the accuracy of the basis set by allowing a more flexible description of the radial behavior of orbitals.

The gaussian functions are grouped in shells, each characterized by the same total angular momentum quantum number L, and shells with different L numbers are used together to form contracted basis functions, where multiple primitive Gaussians are linearly combined with fixed coefficients.

A widely used notation is the Pople notation. A basis set written as $n - ijG$ or $n - ijkG$ indicates that:

- The core shell is formed by n primitive Gaussians contracted to form a single function.

- The valence shell is represented by two (DZ) or three (TZ) functions formed by ij or ijk primitive gaussians, respectively.

A formula like 6-31G indicates that the core shell is represented by a single GTO formed by the contraction of six primitive Gaussians. Each valence orbital is described by two functions: one GTO resulting by the contraction of three Gaussian primitives, and another GTO composed of a single, uncontracted Gaussian primitive.

In 6-31G, the single GTO representing the core shell ϕ_{core} can be written as

$$\phi_{core} = \sum_{i=1}^{6} c_i \, e^{-\alpha r^2} \tag{2.22}$$

And the valence orbitals $\phi_{valence}$ have the form:

$$\phi_{valence} = c_1 \sum_{i=1}^{3} a_i \, e^{-\alpha_i r^2} + c_2 e^{-\beta r^2} \tag{2.23}$$

Where the first term is a contracted triple-primitive Gaussian GTO (flexible main shape), and the second term is a single primitive GTO (captures tail behavior and polarization).

To improve the treatment of valence electron behavior — especially in systems involving anions, weak bonds, or diffuse charge distributions — diffuse functions are added to the basis set. This is denoted by a "+" sign (e.g., 6-31+G), indicating the inclusion of Gaussian functions with small exponents that are very spread out in space. 6-31+G, for example, a diffuse s and p function is added to non-hydrogen atoms, while in 6-31++G, diffuse functions are added to hydrogen as well. Finally, polarization functions can be included to allow for more accurate modeling of non-uniform charge distributions and anisotropic bonding. These are indicated by an asterisk (e.g., 6-31G* or 6-31G**) and involve adding functions with angular momentum one unit higher than required for the atom in question, e.g., d-type orbitals, d-functions, on second-row elements or p-functions on hydrogen. In In 6-31G*, a d-function is added to non-hydrogen atoms; in 6-31G**, a d-function is added to heavy atoms, and a p-function to hydrogen.

2.4.2 Pseudopotentials

During a chemical reaction, the innermost orbitals do not often change significantly. Therefore, they can be treated as an average potential, to reduce the calculation time without compromising the quality of the results. To achieve this, the Effective Core Potentials, ECP were developed.[16–18] ECPs are single-electron operators that replace the explicit treatment of the core electrons, effectively describing the interaction between the core and the valence electrons. ECPs are defined by the parameters of the following expansion:

$$ECP(r) = \sum_{i=1}^{M} d_i r^{n_i} e^{-\zeta_i r^2} \qquad 2.24$$

Where:

- M is the number of terms in the expansion.
- d_i is a coefficient of the i-th term.
- r is the distance from the nucleus. n_i is the power of r of the i-th term.
- ζ_i is the exponent of the Gaussian for the i-th term.

Each ECP is characterized by a specific set of d_i, n_i, and ζ_i. ECPs can describe both a small core, the average orbitals are only those of the innermost shell, and a large core, in which the orbitals of the valence shells are also included in the approximation.

2.5 The Cluster Model

For the discussion of the surfaces of the alkaline-earth oxides under study, the cluster model was adopted and preferred to periodic calculations. This choice was dictated mainly since many physical phenomena such as the interaction of a catalytic site involve species well located on the surface. It is therefore possible to describe the substrate using a cluster formed by a finite number of atoms or ions to reproduce the site of interest. However, a cluster simply shaped in this manner is not a good representation of reality, since unrealistic effects are created at its edges. Such unrealistic effects derive from missing the calculation of the exchange repulsion between ions or edge atoms and the surroundings of the cluster, and from the absence of the long-range electrostatic contribution deriving from the infinite surface.

The latter factor is very important in the ionic solids, in which the Madelung field plays a fundamental role in the description of the potential. Usually, the embedding method is used: the cluster is immersed in surroundings that depends a lot on the type of solid investigated. In this case, the embedding is composed of punctual charges (PC) of value +2, so to have a faithful reproduction of the Madelung potential on the adsorption site in examination.

For example, if we consider a pair of ions separated by a distance r, the attractive electrostatic energy between the two ions is given by Coulomb's law:

$$E = \frac{Z^+Z^-e^2}{4\pi\epsilon_0 r} \tag{2.25}$$

But when a crystal is studied, there are way more interactions to take into consideration. In face centered cubic (FCC) lattices, as in the case of oxides of alkaline-earth metals, one ion feels attractive energy from the six first neighbors, repulsive from twelve seconds, and so on. The electrostatic energy of a couple of ions inside an ionic crystal therefore has the form:

$$E = A\frac{Z^+Z^-e^2}{4\pi\epsilon_0 r} \tag{2.26}$$

Where *A* is the Madelung constant. It depends solely on the geometry of the lattice, and it is a sum of terms that depends on the number of subsequent neighbors and their distance from the atom taken into consideration. To consider the Madelung field, various embedding techniques were developed, ranging from Green functions to empirical methods to describe the surroundings of the cluster.

In the case of ionic crystals, an accurate description of the external long-range electrostatic potential generated by the infinite crystal is required.

In this regard, external potentials $V_{ext}(r)$, defined as sum of contributions $V_i(r)$ are used, centered on the atoms R_i of the substrate:

$$V_{ext}(r) = \sum_i v_i(r - R_i) \tag{2.27}$$

$v_i$is usually treated as a simple punctual charge, and the different embedding schemes differ only in the way in which the integrals $\langle\mu(r)|V_{ext}(r)|\nu(r)\rangle$ are calculated.

However, the use of external potential involves a series of problems. For example, the interruption of the charge flow to the edges of the cluster is not compensated in any way, and the absence of the Pauli repulsion between the electrons of the ions at the edges of the cluster and the punctual charges results in a high polarization that cannot be found in nature. Such polarization is particularly relevant in the case of anions as they have an electronic cloud that is more extended. Consequently, the wave function of the cluster extends beyond the size of the cluster itself. Moreover, this "unnatural" polarization does not allow to relax the degrees of freedom between clusters and PCs, since the two would collapse.

To avoid the artificial polarization of the anions O^{2-} on the edge of the cluster, the positive charges at the interface were replaced with effective core potentials (ECP), without their associated wave functions that provide a representation of the cation's finite dimension. With the ECPs, the repulsion of exchange between the electrons of the oxide ions and the cations is solved. The cations of interface treated in this way are denoted later as M^{*}.

2.6 Calculation Details

In this work, the interaction of BF_3 with several alkaline-earth metal oxides sites was modeled using clusters surrounded by punctual charges ± 2.

For all oxides, terrace sites (100) centered on one oxygen were considered. The stoichiometry of these clusters is the same for all terraces: $\mathrm{M_9O_9M_{16}^{*}}$, Figure 2.1.

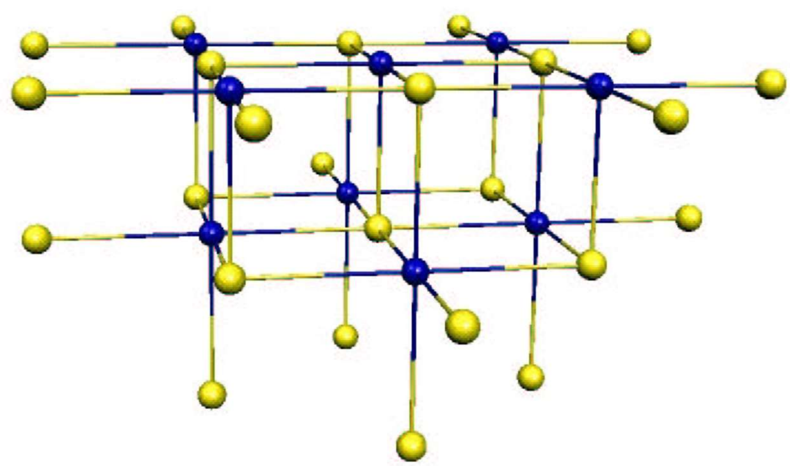

Figure 2.1 - Terrace model (100); In blue are oxygen atoms, in yellow the atoms of the metal

For MgO and CaO, low coordination (LC) anions were also considered, in edges, steps, and vertices. The corresponding clusters are $\mathrm{M_{10}O_{10}M_{14}^{*}}$, (edges), $\mathrm{M_{11}O_{11}M_{13}^{*}}$ (steps), $\mathrm{M_4O_7M_9^{*}}$ (vertices), Figure 2.2, Figure 2.3, and Figure 2.4, respectively.

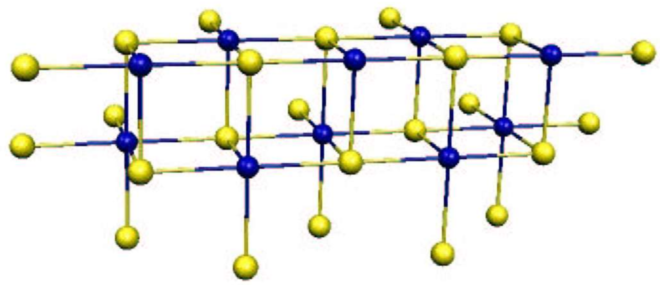

Figure 2.2 - Edge model.

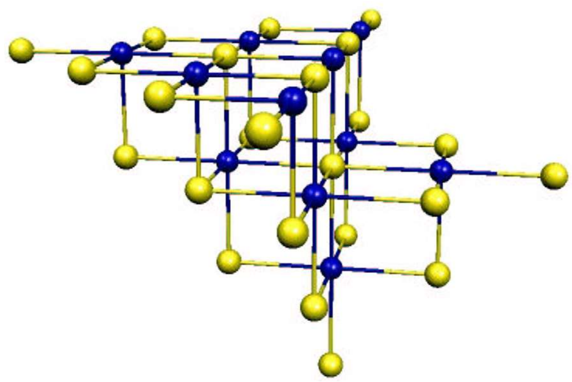

Figure 2.3 - Step model.

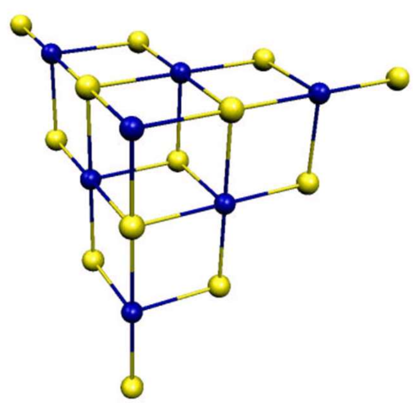

Figure 2.4 - Vertex model.

On terrace sites, only the central oxygen and the adsorbed BF_3 molecule were left to relax. For edge sites, the four and three nearby cations were also included.

The base set 6-31+G* was used for the B and F atoms.[19] On all oxides, oxygen ions were described with a base 6-31G; oxide anions where the BF_3 is bound (adsorption site) were described with the base 6-31+G*, more flexible. The cations Mg^{2+} were treated with a 6-31G base [20]except the first ones close to the adsorption site where a 6-31G* base was used. The Ca^{2+}, Sr^{2+} and Ba^{2+} cations were treated with a base lanl2dz[16–18] associated with the ECP lanl2, which describes a small core. The basic set of the cations closest to the BF_3 was increased with a polarization function *d* with exponents respectively α_{Ca} = 0.216, α_{Sr} = 0.245, α_{Ba} = 0.275. The cations at the interface between the quantum cluster and the block of punctual charges had a Stevens, Basch, Krauss–type pseudopotential,[21] For Mg^{2+}, Ca^{2+}, and Sr^{2+} ions, and a lanl1-type ECP for Ba^{2+}, to describe a large core.

The BF_3 adsorption on oxide surfaces was compared with the interaction of BF_3 with a series of neutral molecules with donor properties: NH_3, H_20, $(CH_3)_2O$, CH_3OH, $P(CH_3)_3$ and with different charged species: OH^-, CN^-, F^- and H^-. The structure of the correspondent, X-BF3 complex, where X stands for the donor molecule was fully optimized through analytical gradients, without symmetry constraints. The molecules involved in the complexes with BF_3; were treated using a 6-31+G* base, except for the hydride anion and the resulting molecular complex, for which a 6-31+g ** base was preferred, since it is the only one with a reasonable H^- ionization potential (IP) value.

All calculations were performed through the Gaussian-98 software.[22]

3. Results

In this work, some geometric parameters related to BF_3 were taken into consideration. The variation of such parameters is an indication of a partial charge donation from the basic site to BF_3. In all the complexes considered, there is immediately an important geometric distortion of BF_3, whose structure goes from trigonal planar to tetrahedral. The bond angle α(XBF) was investigated, as its variation is an indication of how much the BF_3 closes when it forms the complexes. These values have been taken like average of the three corners α(XBF) of the complex.

Another important parameter is the B–F distance r(B–F). The bond with a strong donor species has the result of the partial population of an antibonding B–F orbital, with the consequent extension of the distance B–F. In BF_3, the calculated value of r(B–F) is 1,322 Å, while the experimental value is 1,312 Å.[23]

in the X-BF_3 complexes, the increase of provides a measure of extension of the charge donated from X to BF3. The values of r(B–F) reported later, were calculated as an average of the three B-F distances.

Another geometric parameter considered was the B-X distance, r(B–X). It is expected that a stronger bond, with a pronounced donor character, will result in a shorter distance. Surely, this type of comparison is fully justified when bonds containing oxygen are considered, since the atomic dimension of the donor is the same. The comparison was meaningless when bonds containing N or P were considered, or the anions, because of their different atomic dimensions.

The geometric structure of the coordinated BF_3 probe molecule is not the only indication of the charge-donation from the binder. Changes in vibrational frequencies of the B-F bond stretch are directly related to the changes in the corners and in the bond lengths. In our calculations this quantity is reproduced for the free molecule. A similar accuracy is expected for molecular complexes. Clearly, since the coordination of a donor species implies an extension of the B-F bond, we expect that the corresponding stretching frequency will decrease. the B–F vibration frequency is important since it is experimentally accessible through IR or Electron Energy Loss spectroscopies (EELS). The vibrational frequencies were calculated by calculating the second derivative of the total energy with respect to the internal coordinates. As result, the stretching frequencies of the B–F bond in the free molecule were ν_s = 874 cm^{-1} and ν_{as} = 1438 cm^{-1}. The experimental values are ν_s = 888 cm^{-1} and ν_{as} = 1436 cm^{-1}, respectively.[24] Given the substantial similarity of the two values, no scale factor was applied to the calculated frequencies.

Other quantities considered were the bond energies of the core levels of B (1s) and F (1s) of the adsorbed BF_3 molecules and the 1s level of the oxide anion to which they are linked. In general, shifts of core levels compared to separate and non-interacting units, oxide and BF_3, provide a qualitative, non -quantitative, measure of charge flux from the surface to the adsorbate, therefore of the degree of surface basicity. In fact, many effects contribute to the final shift and that this cannot be related simply to the amount of charge transferred.

The energy of a level (e.g., core) is the result of contributions due to effects of the initial and final states.[25] The initial state effects are those that derive from the structure electronics determined for the neutral system: a wave function for the state effects ψ_i^{n-1} with $n-1$ electrons, it is obtained by annihilating the i-th orbital from the neutral self-consistent field (SCF) wavefunction with n electrons $\psi_i^{n-1} = a_i^\dagger\, \psi_{SCF}^n$, where $a_i^\dagger$ is the annihilation operator of the i-th SCF orbital. The remaining orbitals are not relaxed to shield the hole left by ionization. The corresponding energy is the one of the ionized orbitals, $-\epsilon_i$. To determine the final state effects, the SCF function ψ_{SCF}^{n-1} for the final state with $n-1$ electrons must be considered, by relaxing the electrons in the presence of the hole, but without this being occupied by an electron from the external shell. The energy of ψ_{SCF}^{n-1} is lower than the energy of $a_i^\dagger\, \psi_{SCF}^n$, which considers the initial state effects, and it is called relaxation energy. The energy of the final state, which considers relaxation, is given by the difference in energy between the initial and final states $E[\psi_{SCF}^{n-1}] - E[\,\psi_{SCF}^n]$. In this work, the changes of the Kohn-Sham eigenvalues $-\epsilon_i$ were considered as a measure of the core levels energy shifts that take place during the adsorption. in this way, the effects of the final state are not considered. The quantity is reported as $\Delta\epsilon_{1s}(F)$, $\Delta\epsilon_{1s}(B)$, $\Delta\epsilon_{1s}(O)$, where a positive (negative) difference is assumed where the shift is towards smaller (larger) energy levels of core.

Since we are interested in relative displacements for a series of systems, the approximation to consider the effects of the initial state is justified, and it provides a consistent internal measure of the core level shifts.

Lastly, we considered the bond energy of the BF_3 with the binder. The bond energy could be less indicative of the type of bond and the degree of basicity of the binder. In fact, other terms could contribute to the bond energy, such as dipole-dipole interactions, polarizations intra-unity, and so on, not directly connected with the amount of charge transferred. However, the experimental measure of absorption energy, from TDS spectroscopy was used to quantify the extension of the surface basicity, and a theoretical study of the potential correlation between the oxide basicity and the bond energy becomes relevant.

The observables described above are indirect indicators of the number of electrons donated by the binder and they must be correlated with another term that directly measures the basicity of the donor atom. Here, two quantities, somehow interconnected, can be used. A stronger basicity corresponds to higher energy levels of the donor atom. On the contrary, if the donor levels are deep in energy, their tendency to lose electronic charge is modest. The position of the donor levels can be derived from the simple energy value of the highest occupied Kohn-Sham level (HOMO) or, alternatively, from its vertical ionization potential (IP). The latter is a measurable quantity for both molecular binders and surface oxide sites. However, the trends are very similar when the HOMO energies are correlated to the geometric and electronic indicators of the ligand basicity. Sometimes, the correlation is even better than the one with the values of the ionization potential.

3.1 Molecular Complexes

Figure 3.1 shows the geometric structures of the molecular complexes between the BF_3 and small neutral molecules with donating properties (basic).

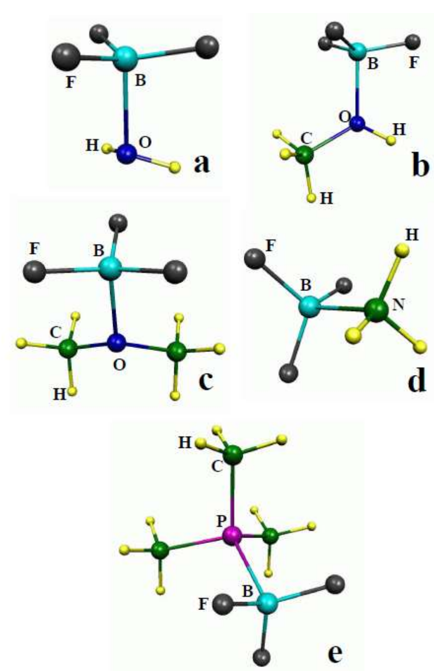

Figure 3.1 - Molecular complexes of neutral molecules with BF_3: (a) $H_2O–BF_3$; (b) $CH_3OH–BF_3$; (c) $(CH_3)_3O–BF_3$; (d) $NH_3–BF_3$; (e) $P(CH_3)_3–BF_3$

In Figure 3.2, the complexes between BF_3 and some anionic species are shown.

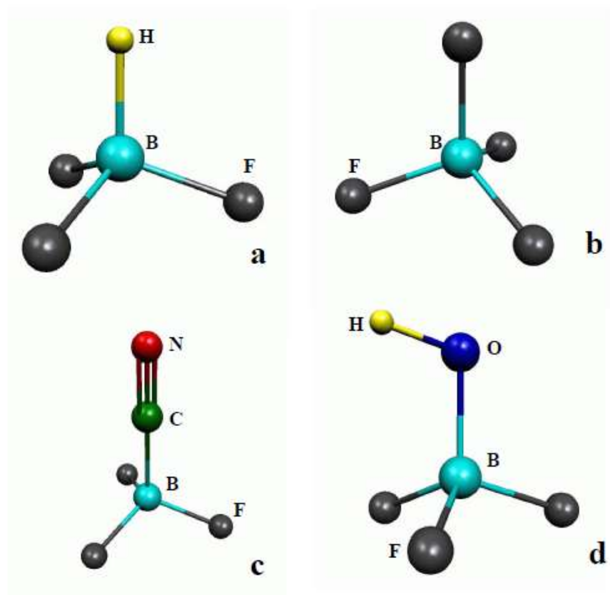

Figure 3.2 - molecular complexes of anions with BF_3: (a) H^-–BF_3; (b) F^-–BF_3; (c) CN^-–BF_3; (d) OH^-–BF_3

In Table 3.1, the calculated geometric properties (Å for the distances and ° for the angles) of the X–BF_3 complexes, together with the ionization potential (eV), the HOMO energy (eV), the B–F symmetric and asymmetric stretching frequencies (cm^{-1}) and the intensity I (Km mol^{-1}) of the asymmetric mode are shown.

Table 3.1 – Geometric properties od moleculart complexes X–BF_3

	IP	E_{HOMO}	r(B-X)	r(B-F)	$\boldsymbol{\alpha}$(XBF)	$\boldsymbol{\nu_s}$	$\boldsymbol{\nu_{as}}$	I($\boldsymbol{\nu_{as}}$)
	eV	eV	Å	Å	°	cm^{-1}	cm^{-1}	Km mol^{-1}
BF_3	15.29	11.98	–	1.322	–	874	1438	469
H_2O–BF_3	12.70	8.69	1.815	1.355	100.3	835	1319	467
CH_3OH–BF_3	10.89	7.67	1.707	1.365	102.1	827	1239	410
$(CH_3)_3O$–BF_3	9.96	7.17	1.694	1.368	102.9	806	1268	396
NH_3–BF_3	10.83	7.29	1.686	1.380	104.4	873	1226	425
$P(CH_3)_3$–BF_3	8.35	6.18	2.079	1.387	105.0	838	1151	266
CN^-–BF_3	4.04	–	1.644	1.411	108.8	669	1064	377
F^-–BF_3	3.52	–	1.418	1.418	109.4	741	1053	455
OH^-–BF_3	1.72	–	1.453	1.430	110.5	735	1172	243
H^-–BF_3	0.21	–	1.229	1.432	110.6	914	902	191

in Table 3.2, the shift of the B, F, O 1s orbitals $\Delta\epsilon_{1s}$ (eV) and the bond energy D_e (eV) between BF_3 and the molecules are shown.

Table 3.2 – B, F, O 1s orbitals shifts and bond energy in the molecular complexes.

	$\Delta\epsilon_{1s}(\mathrm{F})$ eV	$\Delta\epsilon_{1s}(\mathrm{B})$ eV	$\Delta\epsilon_{1s}(\mathrm{O})$ eV	D_e eV
H_2O–BF_3	1.47	0.98	-2.67	0.42
CH_3OH–BF_3	1.84	7.671.36	-2.14	0.55
$(CH_3)_3O$–BF_3	2.00	7.171.51	-2.57	0.56
NH_3–BF_3	1.22	7.291.85	–	1.02
$P(CH_3)_3$–BF_3	2.49	6.182.45	–	0.61
CN^-–BF_3	6.92	6.86	–	2.30
F^-–BF_3	7.72	7.26	–	3.50
OH^-–BF_3	7.99	7.84	–	3.86
H^-–BF_3	8.89	8.93	–	4.01

The characterization of geometric and electronic properties of BF_3 through small molecules is of great use, since such small molecules have a well-known basicity, obtained from acid-basic reactions, and a comparison between the basic sites studied here and these molecules immediately makes possible the ranking of the basic sites in a hypothetical basicity scale. It is noted that oxygen-containing bonds form relatively weak bonds with BF, roughly half of an eV.[9] Water has the largest ionization potential and the H_20–BF_3 complex shows the greatest B–F distance, 1,815 À, Table 3.1, and the smallest shift in the stretching frequency and in the energies of the F (1s) and B (1s) core levels, Table 3.2. By replacing a hydrogen atom with a methyl group, CH_3OH, the B–O distance decreases to 1,707 À and the shift in vibrational frequencies and the core levels become more pronounced. In the complex with dimethyl ether, $(CH_3)_2O$–BF_3, all the indicators show a higher basicity of the molecule, consistent with an IP approx. 2 eV smaller than the one of water. It is noticeable that the angle α(XBF) changes very little with the binder.

Two binders with different basic properties, ammonia and trimethyl phosphine were also considered. NH_3 has an ionization potential like the one of CH_3OH and similar geometric indicators of the complex with BF_3 too, Table 3.1. On the other hand, the shifts of core levels are slightly different, being larger for the boron than for the fluorine, and they change in the binders containing oxygen, where higher shifts are found for the 1s level of the fluorine, Table 3.2. The bond energy is also greater, almost double that for binders containing oxygen. $P(CH_3)_3$ has the minor ionization potential between the neutral molecules considered, and in fact the longest B–F bond, the smallest asymmetrical stretching ν_{as}, and the greatest shift of gold and fluorine core levels are observed. This suggests a stronger basic character of $P(CH_3)_3$ but no direct correlation with the bond energy, only 0.61 eV, is noticed. The $P(CH_3)_3$–BF3 complex was characterized experimentally[26] and theoretically with Hartree-Fock calculations.[27] The experimental bond energy 0.82 eV, is slightly higher than that calculated here probably for the presence of dispersive contributions in the bond that are not properly described in DFT.

A special analysis is required for the 1s core level shift 1s oxygen atom to which the BF_3 is linked. The presence of a charge flux from the donor molecule to the acid BF_3 fragment is expected to result in a stabilization of the O 1s level (negative shift, $\Delta\epsilon < 0$). This is in fact what it is found, Table 3.2. The shift is large, more than 2 eV, and in the right direction, but as the other quantities considered so far, there is not a simple correlation with the basicity of the X molecule. The water, which is the weakest base considered, exhibits the largest $\Delta\epsilon_{1s}(O)$, 2.7 eV; methanol and dimethyl ether, clearly stronger bases, show smaller shifts, Table 3.2. This suggests that the position of the 1s level of the O is very sensitive to other electronic changes and it is not simply correlated to the amount of charge transferred. This result anticipates the conclusion about the use of $\Delta\epsilon_{1s}(O)$ as a measure of the surface basicity of the oxide surfaces.

We now consider the formation of BF_3 complexes with some stable anions, CN^-, F^-, OH^-, H^-, all characterized by positive electronic affinities of the corresponding neutral units. First, we notice the greatest strength of the bond between the anions and the BF_3, with respect to the neutral bases, 2.30 eV for the CN^-, 3.50 eV for F^-, 3.86 eV for OH^- and 4.01 eV for H^-, Table 3.2. This strong bond is accompanied by a big change in the length of the B–F bond, which becomes larger than 1.4 Å, and by a corresponding shift of the B-F stretching frequency which becomes smaller of 1200 cm^{-1}. It is not even surprising the Shift of the core levels of the orbital 1s of B and F which is greater than 6 eV. It could be objected that while the reference is the neutral molecule BF_3 the complex has a net negative charge, with consequent shift of all levels of core towards lower bond energies, without relation with the extension of the transferred charge.

3.2 Surface Complexes

In this section, the potential use of BF_3 as a probe of the basicity of alkaline-earth or of sites differently coordinated on the surface of the same oxides is considered. For each case, the geometry of the absorbent molecule was fully optimized, and the corresponding properties were determined.

In Table 3.3, the geometric properties of the surface complexes with BF_3 is shown.

Table 3.3 – Geometric properties of the surface complexes with BF_3

	IP	E_{HOMO}	r(B-X)	r(B-F)	$\boldsymbol{\alpha}$(XBF)	$\boldsymbol{\nu_s}$	$\boldsymbol{\nu_{as}}$	I($\boldsymbol{\nu_{as}}$)
	eV	eV	Å	Å	°	cm^{-1}	cm^{-1}	Km mol^{-1}
$MgO_{terrace}$	6.71	4.89	1.489	1.415	109.3	771	1118	366
$CaO_{terrace}$	4.93	3.36	1.449	1.431	111.1	764	1041	378
$SrO_{terrace}$	4.45	2.91	1.430	1.440	112.1	761	997	341
$BaO_{terrace}$	3.42	2.15	1.421	1.445	112.6	751	973	313
MgO_{edge}	6.50	4.94	1.409	1.439	109.3	805	1030	318
MgO_{step}	5.68	4.07	1.408	1.446	110.6	782	949	380
MgO_{corner}	4.57	2.96	1.387	1.467	111.8	806	1042	312
CaO_{edge}	4.89	3.49	1.391	1.454	111.8	775	974	428
CaO_{step}	3.95	2.52	1.389	1.461	112.1	755	909	394
CaO_{corner}	3.60	2.24	1.334	1.434	110.4	736	1012	334

In Table 3.4, the B, F, O 1s orbitals shifts and bond energy in the surface complexes with BF_3 is shown.

Table 3.4 – B, F, O 1s orbitals shifts and bond energy in the surface complexes with BF_3.

	$\Delta\epsilon_{1s}(\mathrm{F})$ eV	$\Delta\epsilon_{1s}(\mathrm{B})$ eV	$\Delta\epsilon_{1s}(\mathrm{O})$ eV	D_e eV
$MgO_{terrace}$	3.23	3.07	-2.71	2.13
$CaO_{terrace}$	3.86	3.84	-3.22	3.17
$SrO_{terrace}$	3.98	4.06	-3.16	3.31
$BaO_{terrace}$	4.23	4.34	-2.40	3.46
MgO_{edge}	3.05	3.19	-2.02	3.06
MgO_{step}	3.92	3.95	-1.75	3.33
MgO_{corner}	4.54	4.74	-1.79	3.88
CaO_{edge}	3.57	1.82	-2.47	4.22
CaO_{step}	4.42	4.64	-2.18	4.52
CaO_{corner}	5.00	5.17	-2.36	4.67

A first observation is that all indicators clearly show the basicity of the oxygen anion of alkaline-earth oxides higher compared to the one with the binders $(CH_3)_2O$, NH_3 or $P(CH_3)_3$, as shown in Table 3.3 and Table 3.4. It is noted that:

- r(B-F) is much longer than in neutral molecular complexes, being greater than 1.4 Å.
- The bond angle α(OBF) is closer to that of a tetrahedral structure, with values of 110.5° ± 1.5°; the frequency of asymmetrical stretching, ν_{as} is 1000-1100 cm^{-1}, while it is 1150-1300 cm^{-1} in molecular complexes.
- The 1s orbital core levels of B and F are shifted by ≈ 3-5 eV towards lower bond energy, while the corresponding shifts in the molecular complexes are ≈1-2 eV.
- The bond energies of the BF_3 with oxide anions are between 2 and 5 eV, i.e., 4 to 5 times larger than in neutral X-BF_3 molecules.

From all this data it emerges quite clearly that the basicity of the anions of alkaline-earth oxides is much more pronounced than that of common basic compounds, such as ammonia, and it approaches the basicity of ionic species such as CN^- and F^-. This is totally consistent with the parameter used to qualitatively measure the basic character of the binder, the vertical ionization potential IP. This quantity in binary oxides is between 3.5 and 6.7 eV, i.e., it is comparable with the IP of the anions, while the same quantity in the considered neutral molecular bonds is between 8.3 and 12.7 eV, Table 3.3 and Table 3.4.

Having established the much more pronounced basic character of the surface anions in the alkaline-earth oxides, the relative differences in basicity by going from lighter alkaline-earth oxides to the heaviest, and from regular to defective sites are considered.

3.2.1 Basicity Trends on MgO, CaO, SrO, BaO Terraces.

The BF_3 was adsorbed on anions oxide penta-coordinated (O_{5c}) on terraces (100) of MgO, CaO, SrO, and BaO, Figure 3.3.

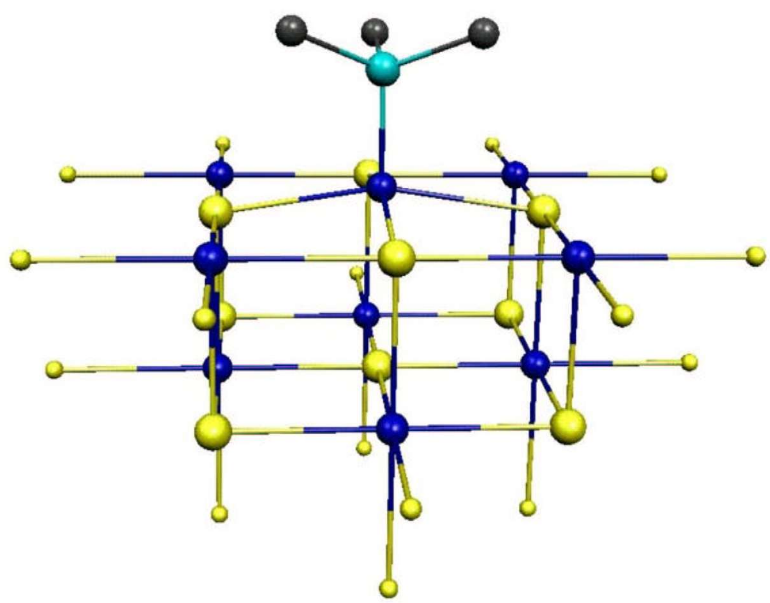

Figure 3.3 - BF_3 adsorbed on a terrace site.

Previous theoretical studies have shown that the surface basicity increases by going down to the series, and this can be easily explained with the variation of the Madelung potential (MP), due to the increase in the lattice parameter.[5] This is reflected in the IP value, that in the MgO, 6.7 eV, is approx. double that in BaO, 3.4 eV, Table 3.3. There is a large variation of the ionization potential, 1.8 eV, going from MgO to CaO, while the difference between CaO and SrO, 0.9 eV, and between SrO and BaO, 0.6 eV, is smaller. A question to be addressed is to what extent this non-monotonous change reflects in the properties of the absorbed BF_3.

The BF_3 binds on the anion O_{5s} of MgO with D_e = 2.13 eV, r(O-B) = 1,489 Å, R(B-F) = 1,415 Å. On the same site of CaO, SrO and BaO there is an increase in D_e, (3.17, 3.36 and 3.46 eV, respectively) and a consequent contraction of r(O-B), 1,449, 1,444, 1,421 Å, respectively. The distance B-F was also slightly elongated in heavier oxides, 1,431, 1,432, 1,445 Å, Table 3.3. Therefore, there is a clear change by going from MgO to CaO, and a less marked variation from CaO to BaO, as expected based on the IP values. This is also reflected on the angle α(OBF), but here the variations, 2 degrees, are probably too small to be detected experimentally. The change in the structural parameters is clearly placed in relation to the increase of donation from oxide anion to BF_3, as shown by the great positive shift of the 1s level of B and F. At the same time, a shift of opposite sign is observed in the 1s level of oxygen, where absorption took place (here the shift is defined as the difference in the energy of the oxygen orbital 1s in the free cluster and in the same cluster with the adsorbed BF_3).

In principle, two observable properties can be used to classify the degree of basicity of the four substrates: the vibrational frequency of the BF_3 measured in IR spectra, and the shift of the core levels measured in XPS spectra. Here, both symmetrical (ν_s) and asymmetrical (ν_{as}) stretching were considered, Table 3.3. However, only ν_{as} seems to provide a practicable tool to measure the degree of charge transfer; in fact, the changes in ν_s are small and do not correlate in a simple way with the degree of surficial basicity. For example, the ν_s values for the adsorbed BF_3 on terrace sites of MgO, CaO, SrO and BaO are grouped around 760 ± 10 cm^{-1} and do not reflect the 3.3 eV IP change going from MgO to BaO. A reason for the absence of correlation of this vibrational mode is that it is strongly coupled with the reticular phonons of the substrate, thus losing the typical local character necessary for an analysis of this type. Compared to this, the ν_{as} frequency is much more useful. In fact, while ν_{as} in MgO/BF_3 is 1118 cm^{-1}, in BaO/BF_3 decreases to 973 cm^{-1}, with CAO and SrO that show intermediate values, Table 3.3. While the frequency change is certainly detectable experimentally, the shifts of the core levels of B and F are not easily measurable in practice, as the free molecule used as a reference is in the gaseous phase. Much more promising would be the possibility of using the 1s core levels of O and their shifts due to adsorption, as a measure of basicity. However, things are not so simple. In fact, we find that the shifts are large and negative, as expected, but do not reflect in a simple way the amount of charge transferred, or the ionization potential of the surface site. On the MgO terrace of, the adsorption of BF_3 leads to a stabilization of the O 1s level of 2.71 eV. On the same site of Cao and SrO, the shift is larger, 3.2-3.3 eV, as expected given the great basicity of these surfaces. Instead, on BaO the shift is only 2.4 eV, Table 3.4. Therefore, the more basic surface does not produce the greatest shift in O 1s orbital. In the past the physical mechanisms that contribute to the final shift in the energies of the core levels of alkaline-earth oxides were analyzed in XPS experiments, and some attempts to correlate this quantity with the ionicity of oxide were proposed.[28] Different mechanisms (Madelung potential, net charge on oxygen, distance O-BF_3, local change of ionicity induced by the absorption of the BF_3, etc.) produce opposite shifts, and the final shift of the O 1s core level can be the result of large terms, but of opposite sign.

3.2.2 Basicity of MgO and CaO low coordination sites

Now let's see the change of basicity going from a highly coordinated site to a low coordination site on the same surface. We considered three defective sites on MgO and CaO surfaces: oxide anions on edges, steps, and vertices. Similar effects are expected on the corresponding sites of SrO and BaO.

In the first two cases, we consider tetra-coordinated oxide anions, O_{4c}, while on the vertex site, a tri-coordinated anion, O_{3c} is present, Figure 3.4, Figure 3.5, and Figure 3.6, respectively. It should be noted that the different surroundings of steps and edges lead to considerably different anions properties. In fact, on MgO_{edge}, the IP = 6.50 eV is only slightly smaller than on the terrace, while on MgO_{step}, the IP 5.68 eV is about 1 eV lower than on the terrace. On the vertex, there is a further reduction of ionization potential, 4.57 eV, Table 3.3. There is also a trend similar for CaO: an anion O_{4c} on the edge exhibits an IP = 4.89 eV very close to the one calculated for the terrace, 4.93 eV. On a step, the ionization potential is about 1 eV lower, 3.95 eV. Finally, on the vertex site, the ionization potential, 3.60 eV, approaches that calculated for the penta-coordinated anion on the BaO surface, Table 3.3. Therefore, based on these quantities, the basic trend can be predicted within a given surface: vertex > step > edge ≈ terrace.

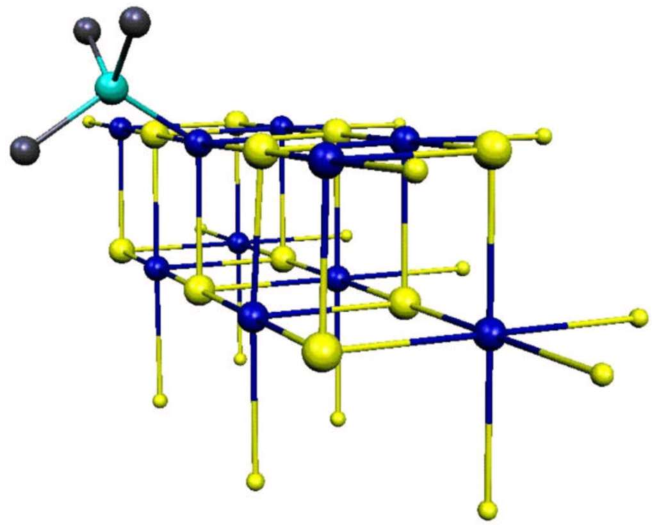

Figure 3.4 - BF_3 adsorbed on an edge site.

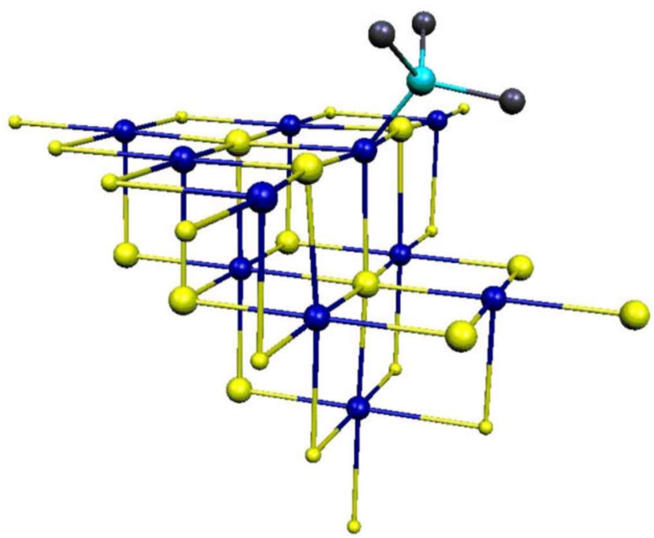

Figure 3.5 - BF_3 adsorbed on a step site.

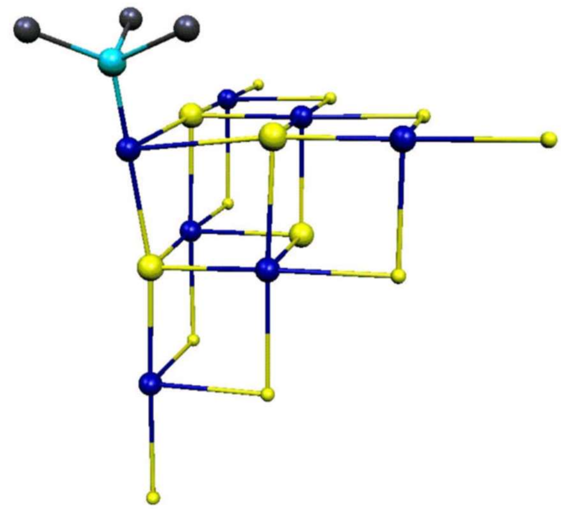

Figure 3.6 - BF_3 adsorbed on a vertex site.

We now consider the properties of the BF_3 absorbed on the low coordination sites of MgO and CaO. In analyzing the geometric properties, it must be kept in mind that the anions oxide with low coordination have a greater possibility of relaxing than those penta-coordinates, and that the steric impediment of the BF_3 which binds to the surface is minor on these sites. This could somehow affect the r(B–O) distance which is significantly less compared to the flat terraces. On MgO_{edge} and MgO_{step}, r(B-O) is 1.41 Å instead of 1.49 Å, the value found for the terrace; On MgO_{corner} it is even less, 1.38 Å, Table 3.2. The same trend is found for CaO, with r(B-0) = 1.39 Å and 1.33 Å respectively for the tetra- and tri-coordinated anions, to be compared with the optimal distance on the terrace website, 1.45 Å. The second consequence is that, while on flat terraces r(B-O) is longer than r(B-F), on low coordination sites we observe an opposite trend, with r(B-F) > r(B-O), Table 3.3. It is not surprising that for the vertex sites of MgO and CaO, the minor r(B-O) distances and the higher bond lengths r(B-F) ledges are obtained. On the other hand, there is no change in the corner α(OBF) going from one site to another, showing that this structural parameter is less sensitive to interaction and does not provide a good measure of the coordination of the ion, Table 3.3.

Absorption on low coordination sites has the result of electronic changes. As with molecular complexes, we first analyze vibrational properties and in particular the shift of the symmetrical and asymmetrical ironing frequencies of the B-F bond. Going from a terrace to a low coordination site, we observe a small blue-Shift of ν_s, but a large red-shift of ν_{as} (from 1118 cm^{-1} to 1030-1040 cm^{-1}, Table 3.3). Therefore, IR spectroscopy should be able to discriminate between a molecule of BF_3 linked to a regular or defective site. in this way, an estimate of the total number of low coordination sites on a given surface could be estimated. In fact, the absorption intensity of the ν_{as} mode is similar on the terrace and on low coordination sites, so that the intensity of the corresponding absorption peaks should be directly related to the number of adsorption sites. There are similar trends for CaO, even if in a less pronounced way. When there is interaction with low coordination sites, ν_s, related to the absorption on terrace sites, remains almost unchanged or is slightly shifted to red, while ν_{as} a red-shift of many tens of cm^{-1}, Table 3.3. The differences between the frequencies obtained on the different low coordination sites (edges, steps, and vertex) are too small to suggest their own use as an indicator of the concentration of each defective site.

The other spectroscopy that could be used to monitor the number and strength of the basic sites is XPS. On the regular surface of MgO, the 1s core levels of B and F are shifted of approx. 3.8 eV towards higher energies, Table 3.4. This shift does not change when the BF_3 is linked to a corner, while it is much greater for the steps, approx. 3.8–3.9 eV, and the vertex, 4.5-4.7 eV. These differences are more than enough to allow peaks identification or shoulders in the XPS spectra, due to adsorption on low coordination sites, but the number of these sites must be sufficiently large to be identified. The same conclusion is valid for the defective sites of CaO. On the terrace, the shift of core levels is approx. 3.8 eV and does not change on the edge site; However, for a step and for a vertex the shift is significantly greater, ≈ 4.5 and ≈ 5 eV, respectively. Therefore, we can conclude that the XPS measures of the shifts of 1s core levels of B e F should be able, at least in principle, to trace and titrate the number of defective sites on the surface through an indirect measure of their charge transfer.

This is not the case for the shift of the oxygen 1s core level. Going from MgO high coordination sites to low coordination sites, the trend is counterintuitive. The shift of the O 1s level induced by the adsorption of the BF_3 on a terrace site, -2.71 eV, it is larger in absolute value than those on the edges, steps and vertices where the shifts are smaller than 2 eV, Table 3.4. Cao exhibits the same trend, Table 3.4. It was shown that the shift of the 1s core levels of or in ionic materials relates to the Madelung potential value on that site.[29] A plausible explanation is that in addition to the adsorption of BF3, the change of coordination, the structural relaxation (therefore the change in Madelung potential) provide important contributions to the final shifts that partially counterbalance those of the increased charge transfer. Therefore, the result for low coordination sites confirms the conclusion that the movement of the O 1s core levels are not indicators of the basicity of the site.

3.2.3 Correlation between the surface basicity and the properties of the adsorbed BF3.

In this section, we analyze the correlation between the surface basicity, measured by vertical IP and the energy of the HOMO (absolute value) of the molecule or oxide ion bonded to BF_3, and the various properties considered as indicators of the amount of charge donated. In doing this, we reported in a single plot the results for molecular binders, for regular oxides and defective sites. The anionic binders were not considered as their accurate treatment would have requested a very large and widespread basic sets. Furthermore, some investigated properties, such as the shifts of core levels, change considerably for charged systems.

Distance B-O

In Figure 3.7and Figure 3.8, the trend of the r(B-O) bond distance versus the ionization potential IP, and versus the HOMO energy, respectively; This is done by including only binders containing oxygen, in order to avoid problems related with the different atomic dimensions of molecules containing P or N.

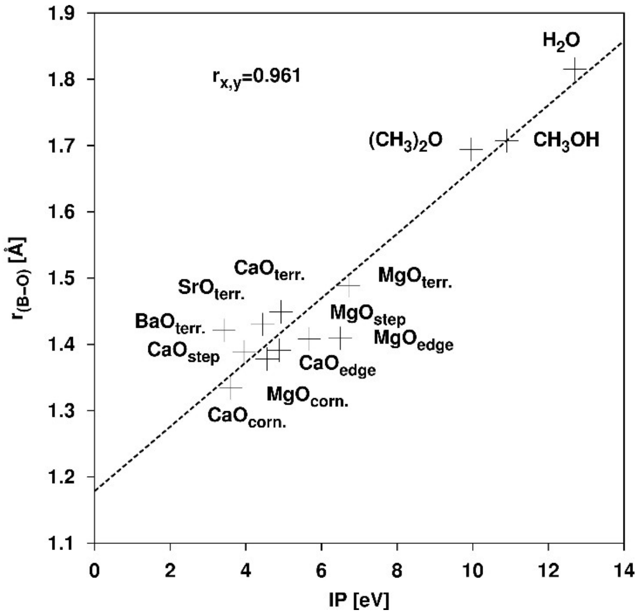

Figure 3.7 - Distance r(B-O) versus to the ionization potential IP.

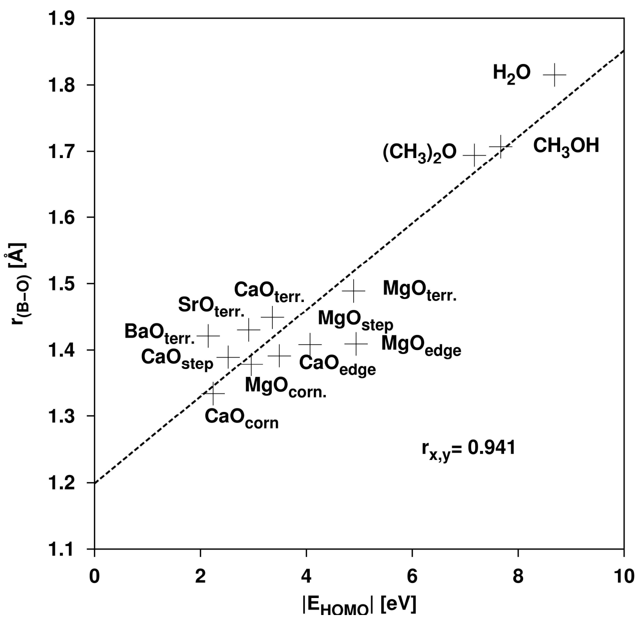

Figure 3.8 - Distance r(B-O) versus E_{HOMO}.

The trend is almost linear, with a correlation coefficient of 0.961 and 0.941, respectively. Oxide anions, more basic, give rise to very minor r(B-O) distances grouped around 1.4-1.5 Å, while the molecular binders produce greater distances. On the MgO and CaO low coordination sites BF_3 forms the short bonds.

Distance B-F

The plots in Figure 3.9 and Figure 3.10 show correlations with r(B-F). Here, all the ligands and the oxide sites were included, and this explains the slightly lower correlation coefficient for IP, compared to the previous case. In general, there is a well-defined increase in r(B-F) with a decrease of IP and E_{HOMO} of the donor group. Also in this case, the vertex sites of MgO and CaO produce the longest r(B-F) distances, while the water is the other extreme of the plots.

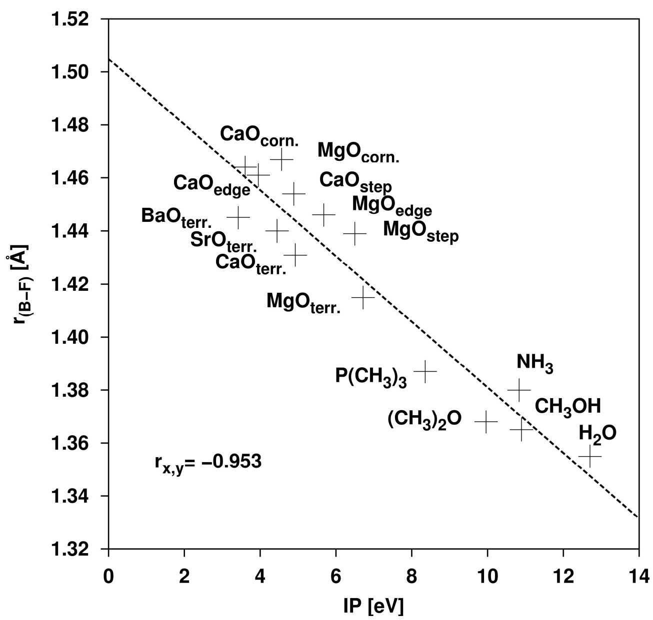

Figure 3.9 - Distance r(B-F) versus the ionization potential IP.

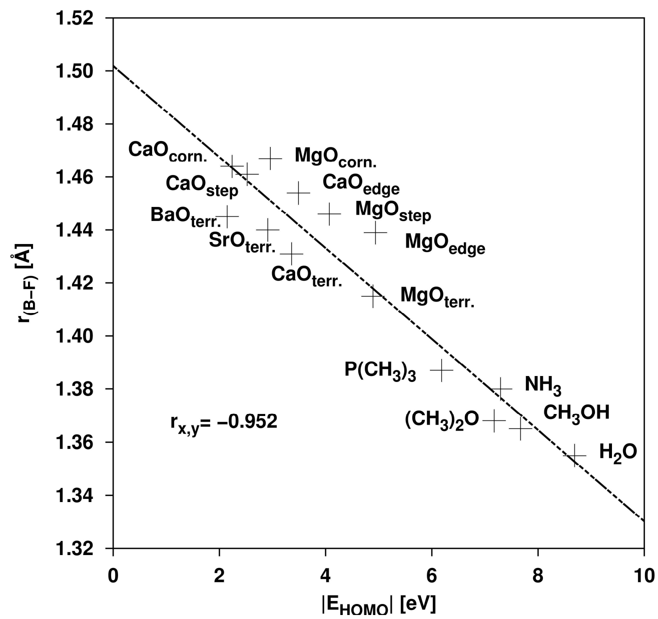

Figure 3.10 - Distance r(B-F) versus E_{HOMO}.

XBF angle

Figure 3.11 and Figure 3.12 show the correlation of the α(XBF) bond angle with IP and E_{HOMO}. In the previous section we mentioned that this quantity is difficult to measure experimentally and therefore gives little help in the characterization of surface basicity. In fact, for example, a 10 eV change in IP of the donor species leads to a change in the α(XBF) angle of 12°, and often the changes are only of 2-3°. However, the α(XBF) angle exhibits a very good correlation in both cases, with the highest correlation coefficient in the case of IP, -0.974, among those considered.

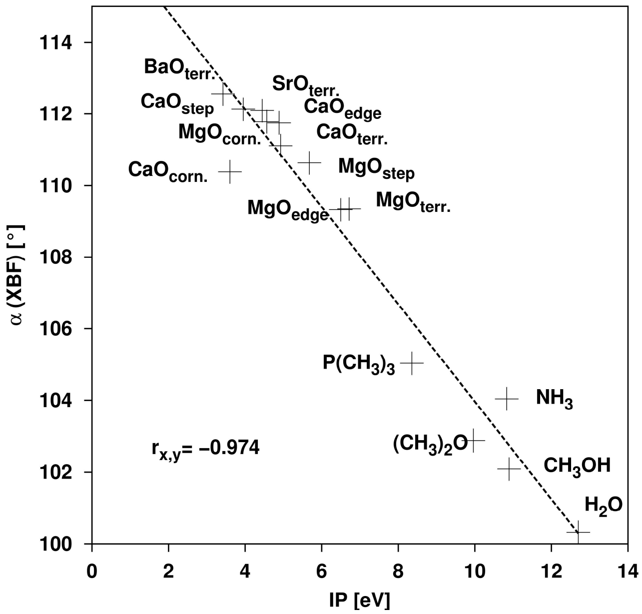

Figure 3.11 - Angle α(XBF) versus IP.

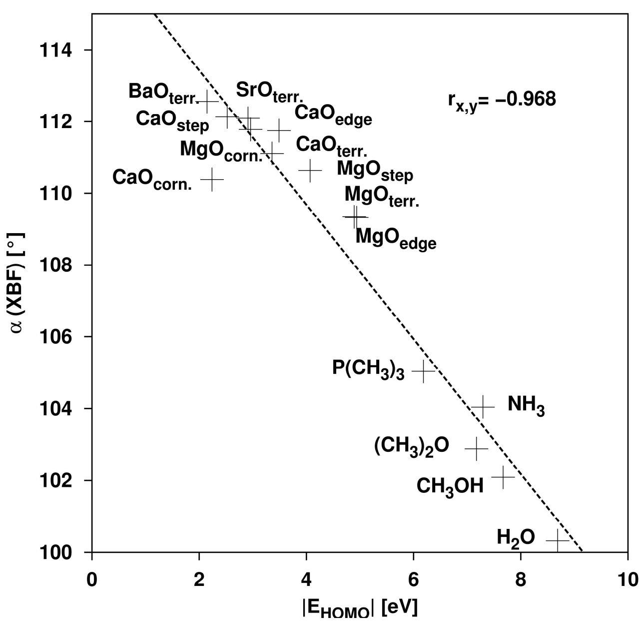

Figure 3.12 - Angle α(XBF) versus E_{HOMO}.

Asymmetrical stretching

In Figure 3.13 and Figure 3.14, the values of the asymmetrical stretching of the B-F bond ν_{as}(B–F) versus IP and E_{HOMO} are shown. This is the most intense and characteristic vibrational mode of BF_3.

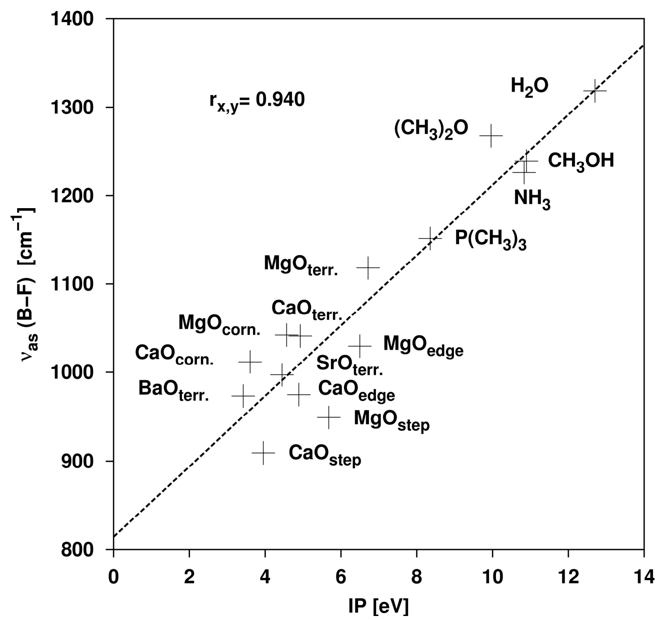

Figure 3.13 – Asymmetric stretching ν_{as}(B–F) versus the ionization potential IP.

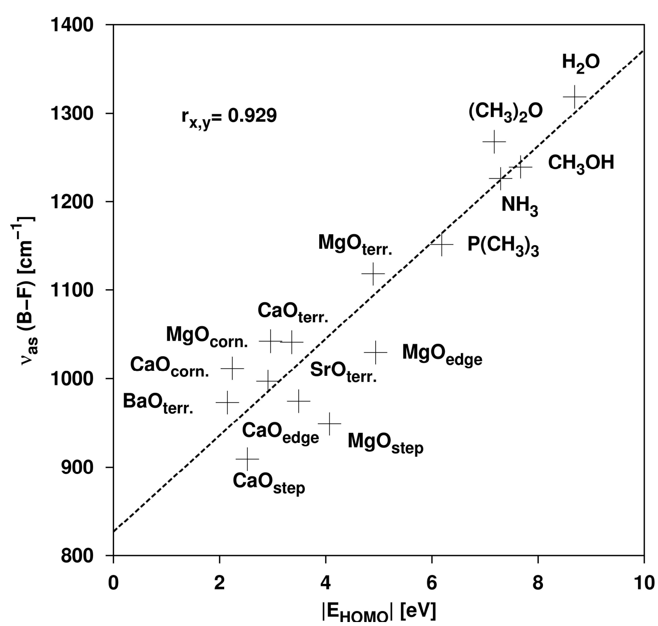

Figure 3.14 – Asymmetric stretching ν_{as}(B–F) versus E_{HOMO}.

The correlation coefficients, 0.940 in the case of IP and 0.929 in the case of the E_{HOMO}, indicate only small variations from linearity. This is an important result as vibrational spectroscopy is easily accessible, and it can in principle provide a valid tool to control the amount of charge transferred to the adsorbed molecules.

1s core level shift of F and B

In the plots from Figure 3.15 to Figure 3.18, the shifts of the 1s levels of F and B are shown. The trends is similar for the two atoms, with a clear tendency towards major shifts when the BF_3 is linked to oxides with respect to molecular binders. Within the oxides, low coordination sites show the greatest shifts in the energy of the 1s core level of B e F. The correlation is greater for boron and for fluorine, both in the case of IP (-0.972 for B, -0.964 for F), both in the case of the E_{HOMO} (-0.980 for B and -0.967 for F). Unfortunately, the shifts of core levels are not easily accessible experimentally, due to the difficulty of measuring the BF_3, gaseous in its free form, to obtain a reference value.

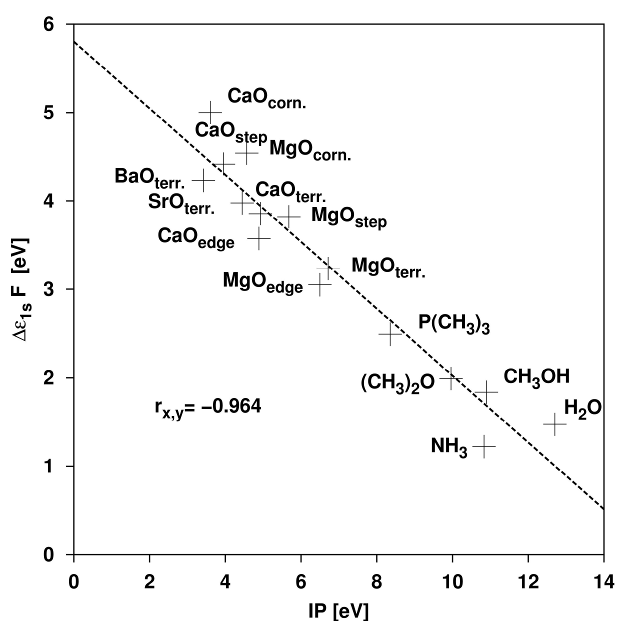

Figure 3.15 – Shift of the fluorine 1s core level $\Delta\epsilon_{1s}$F versus the ionization potential IP.

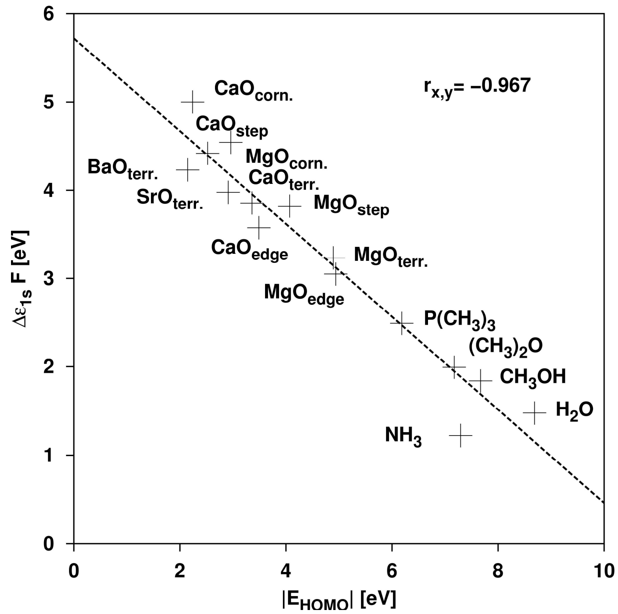

Figure 3.16 – Shift of the fluorine 1s core level $\Delta\epsilon_{1s}$F versus E_{HOMO}.

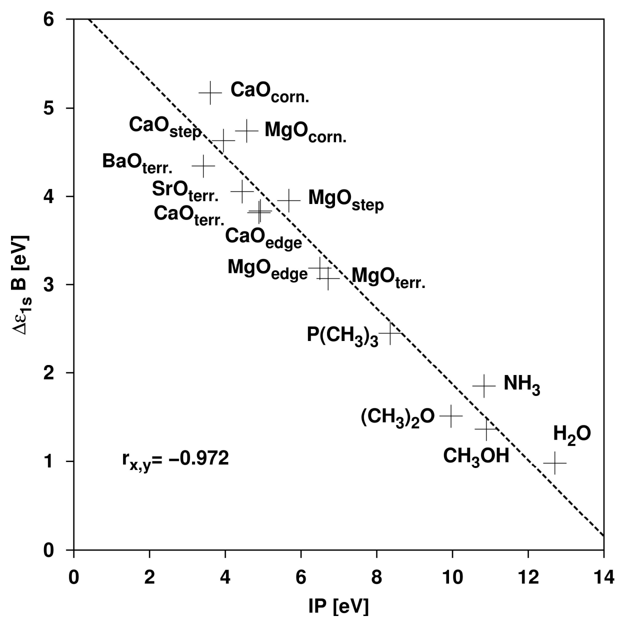

Figure 3.17 – Shift of the bromine 1s core $\Delta\epsilon_{1s}$B versus the ionization potential IP.

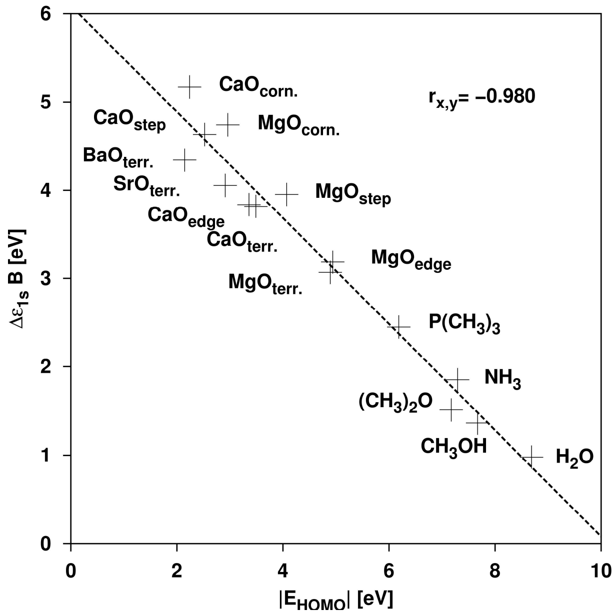

Figure 3.18 – Shift of the bromine 1s core level $\Delta\epsilon_{1s}$B versus E_{HOMO}.

Shift of 1s core levels of O

The analysis of the oxide anion bonded to the BF_3 would be much more interesting, at least in principle, compared to the other shift previously considered. Unfortunately, for the reasons mentioned in the previous paragraph, there is practically no correlation, Figure 3.19 and Figure 3.20. The calculated shifts in the levels of 1s core are completely scattered and do not show relationship nor with IP, nor with E_{HOMO}.

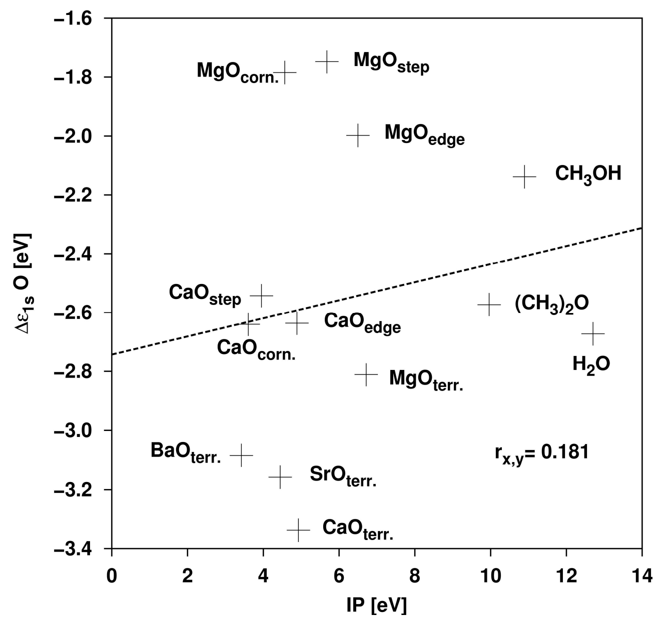

Figure 3.19 – Shift of the oxygen 1s core level $\Delta\epsilon_{1s}$O versus the ionization potential.

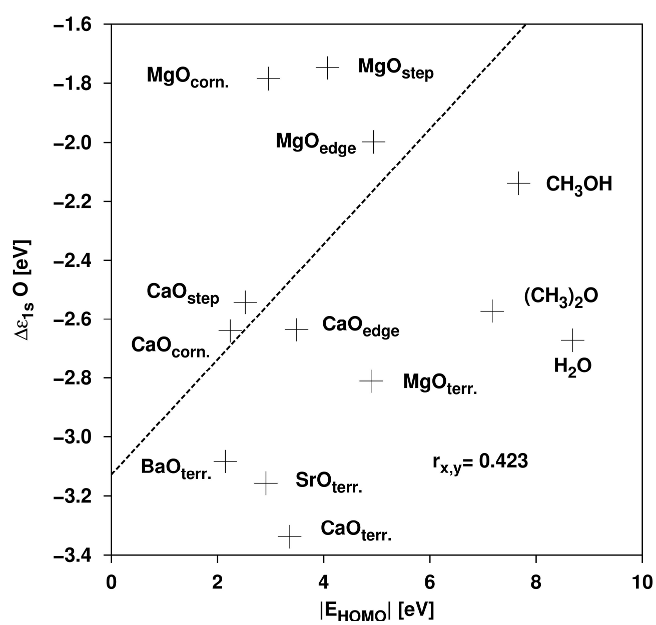

Figure 3.20 – Shift of the oxygen 1s core level $\Delta\epsilon_{1s}$O versus E_{HOMO}.

Binding energy

The last two plots, shown in Figure 3.21 and Figure 3.22, report the values of the bond energy, or BF_3 adsorption energy, versus IP and E_{HOMO}.

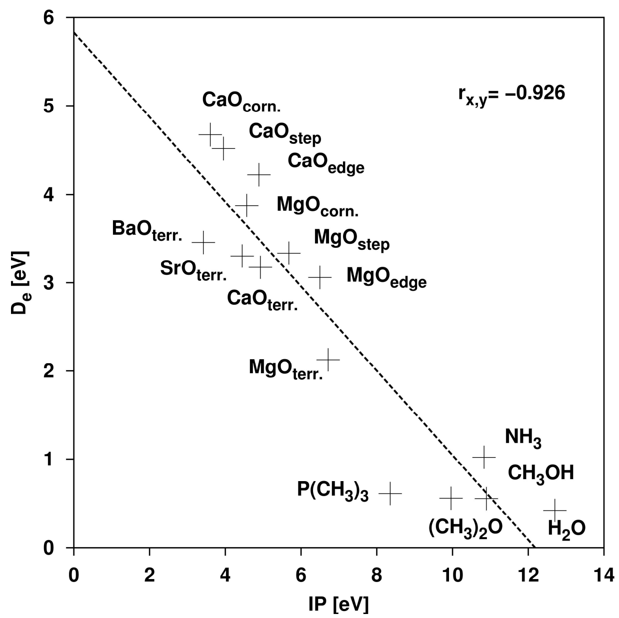

Figure 3.21 – Bond energy D_e versus the ionization potential.

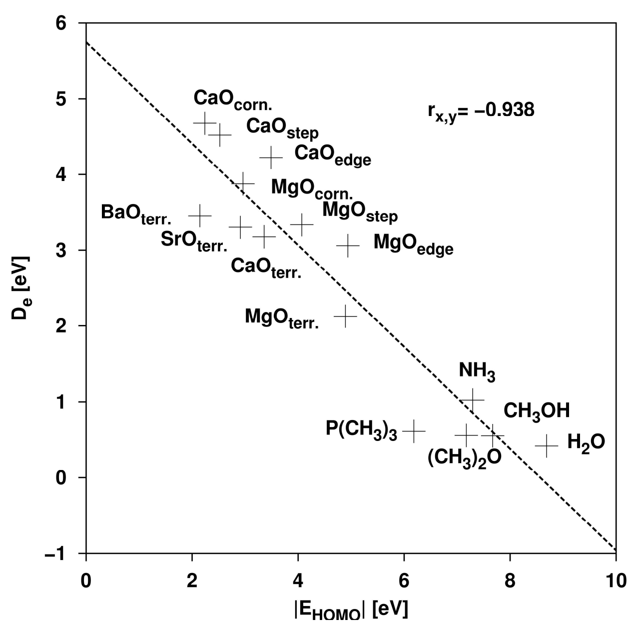

Figure 3.22 – Bond energy D_e versus E_{HOMO}.

There is one direct relationship between the strength of the bond and the basicity of the adsorption site (correlation coefficient -0.926 and -0.938 respectively). Molecular systems only form weak bonds, only $P(CH_3)_3$ deflects partially, forming a slightly stronger link, Table 3.2. The most basic BaO surface binds BF_3 stronger than MgO. The low coordination sites of MgO and CaO create stronger bonds with BF_3 with respect to regular sites. In a TDS experiment, this will result in desorption peaks appearing at higher temperatures. Besides that, the peak area should be proportion to the number of sites involved. A possible issue is that energies of approx. 3-5 eV are necessary to detach BF_3 from MgO and CaO low coordination sites. This results in temperatures well higher than room temperature, with the possible break of the BF_3 molecule before the desorption would occur.

4. The case of Silica and Rutile

During this work it was thought to test the BF_3 adsorption also on an oxide not purely ionic like rutile, TiO_2, and on a covalent oxide such as silica, SiO_2.

4.1 TiO_2

Rutile is a partially ionic oxide: its degree of ionicity is significantly lower than the nominal one, which provides for a charge +4 on the atoms of Ti and a charge -2 on the atoms of O.[30] Thanks to this, a rutile cluster can be both immersed in the surroundings of charges and being terminated by hydrogen atoms. The latter is a common technique to generate clusters of covalent materials such as Si or SiO_2 and that it also works well for TiO_2.[31,32] However, we preferred to model the cluster of rutile by immerging it in a surrounding of charges, assigning a charge +2 to atoms of Ti and -1 to O.[33,34] Such model bonds the BF_3 on the two possible sites of the rutile (100) face: in the two-coordinated oxygen bridging two titanium atoms, and on the tri-coordinated oxygen.

4.1.1 Computational Details

For the simulation of the BF3 bonding the two-coordinated oxygen we used a cluster with a $Ti_4Ca_4O_{16}Ca_{13}^*$ stoichiometry, Figure 4.1.

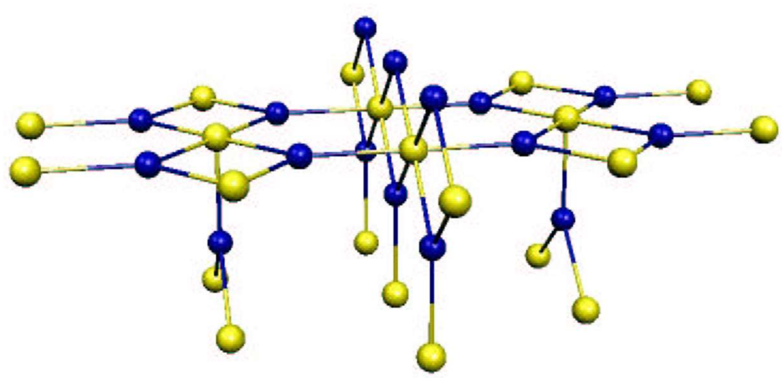

Figure 4.1 – Surface (110) of rutile (space Group $P4_2/mnm$); in blue the oxygen, and in yellow the titanium.

The farthest Ti atoms from the site of interest were replaced with Ca atoms, to have similar atomic dimensions and atomic weight, while maintaining charge neutrality, so as not to load too much the calculations. For the bond on oxygen tri-coordinated we have used a cluster of a $Ti_6Ca_5O_{22}Ca_{14}^*$ stoichiometry. The following bases were adopted: the atoms of Ti were treated with a lanl2dz base, O with a 6-31G, except the BF_3 adsorption site, where a polarization function (*) a diffused function (+). Also, for the BF_3 a 6-31+G* base was preferred. In the case of Ca, those belonging to the quantum cluster (Ca^{2+}) were described with a lanl1mb base, those used as an interface between the atoms of O and the punctual charges have ECPs of type lanl1, suitable for describing a large core. On the TiO_2 surface, both the oxygen atom on which the adsorption took place and the BF_3 molecule were free to relax.

4.1.2 Results

Bridge oxygen

As clearly shown in Figure 4.2, the adsorbed BF_3 on the tri-coordinated O immediately reacts and breaks.

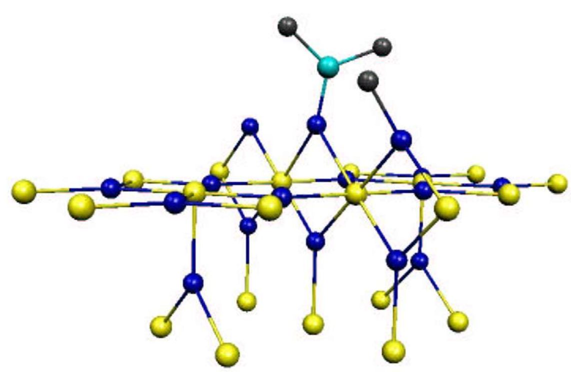

Figure 4.2 – Surface (110) of TiO_2: the fragmentation of BF_3.

This can be due to the low coordination of the bridge oxygens, which are therefore particularly reactive. In this thesis work, the emphasis on the possible reactivity of the BF_3 with the various surfaces was not placed, so the tri-coordinated oxygen was considered, Figure 4.3, to which the data in Table 4.1 are referred.

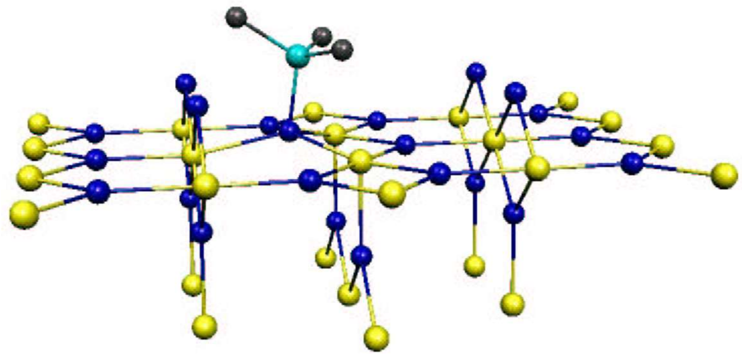

Figure 4.3 – Adsorbed BF_3 by the tri-coordinated oxygen of the TiO_2 (100) surface.

Table 4.1 - Calculated properties of surface complexes between TiO_2, SiO_2, and BF_3.

	IP	E_{HOMO}	r(B-X)	r(B-F)	α(XBF)	$\Delta\epsilon_{1s}(F)$	$\Delta\epsilon_{1s}(B)$	$\Delta\epsilon_{1s}(O)$	D_e
	eV	eV	Å	Å	°	eV	eV	eV	eV
TiO_2	8.19	7.42	1.541	1.407	106.3	0.44	0.13	-0.10	1.18
SiO_2	10.33	8.31	1.665	1.376	104.9	1.74	1.20	-1.77	0.16

Tri-coordinated oxygen

As can be seen from Table 4.1, the geometric properties of the BF_3 adsorbed on the tri-coordinated O of the TiO_2 surface (110) well agree to those obtained in the previous calculations: the distance r(B-F) (1.407 Å) and the bond angle α(OBF) (106.3°) are lower than the respective values considered in the case of oxides surfaces (see Table 3.3), but higher than the respective values related to neutral molecules, Table 3.1; This can be a clue to the not ideal ionicity of rutile. This is also reflected on the ionization potential, whose value (8.19 eV) is much more similar to the ionization potential of neutral molecules than to that of the surfaces (Table 3.1 and Table 3.3); Similarly, E_{HOMO} follows the same trend as the ionization potential: its value (7.42 eV) is between methanol (7.67 eV) and dimethyl-ether (7.29 eV) (Table 3.1). The distance between the BF_3 and the adsorption site, however, is the largest among those considered so far (1,541 Å), and this is consistent with the other calculated parameters.

The problems arise when the electronic properties of the complex are investigated: the shift of the 1s levels of B and F are very close to zero (0.44 and 0.13 eV, respectively), while 4-5 times larger values are expected, in line with the trend previously found (Table 3.4). Even the 1s level of O (-0.10 eV) remains almost unchanged. All this indicates that the cluster model is probably not ideal for modeling the surface of TiO_2, since a potential that does not correspond to reality is generated. In fact, by adding up the charges of the surface, a negative value is obtained (-0.212 a.u.) while the BF_3 bonded to it has a positive value (+0.212 a.u.). Therefore, on this cluster, the BF_3 behaves as a base and not as an acid, as it would be legitimate to suppose. This also explains the reason why we have avoided calculating the vibration frequencies of the B-F bond.

4.2 SiO_2

Silica is one of the most popular minerals in nature and exists in many different forms. The most important crystalline shapes are a quartz α, stable within 573 °C, quartz β (stable from 573 up to 870 °C), γ-tridymite (from 870 up to 1470 °C) and finally, the β -cristobalite, which is stable until the melting temperature of 1713 °C.

The model used refers to quartz α, composed of tetrahedra to whose vertex are the oxygen atoms, and in the center those of silicon. The tetrahedra are linked among them at the vertex, that is, through an oxygen atom that for this reason it is said to *bridge* between two silicon atoms. The Si-O-Si angle is about 144° while the O-Si-O angle (inside a tetrahedron) is 109°. The Si-O bond is approximately 1.60 Å.

4.2.1 Computational Details

To model the adsorption of the BF_3 on the bridge oxygen, a small cluster was used, Figure 4.4.

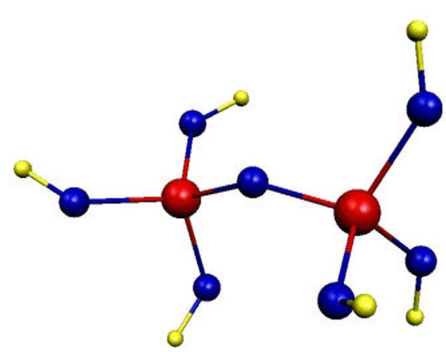

Figure 4.4 – SiO_2 cluster used for calculations: in red the Si atoms, in blue the oxygens, and in yellow the hydrogens.

Here, the dandling bonds were saturated with hydrogen atoms placed along the O-Si crystal directions[35] of stoichiometry $Si_90_7H_6$, in which all the atoms of SI and O, as well as the BF_3 were let relaxing, so to consider only the effects of the local relaxation.[36] The silicon atoms and the BF_3 were described with a 6-314G base while for the Hydrogen was preferred a 6-31G base.

4.2.2 Results

Also in this case, the geometric properties of the complex formed by the SiO_2 cluster and the BF_3, Figure 4.5, correlate well with the respective values previously calculated in the case of alkaline-earth oxides and molecular complexes (see Table 3.1, Table 3.3, and Table 4.1).

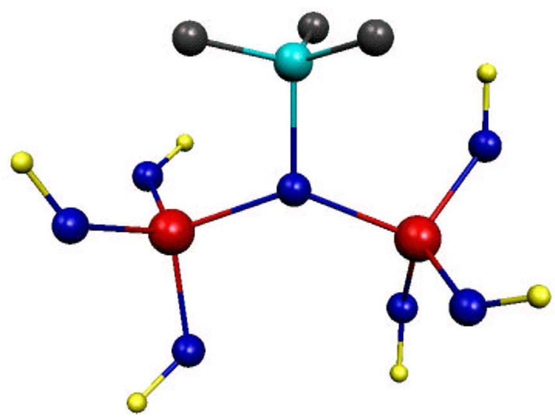

Figure 4.5 – Adsorbed BF_3 on SiO_2 cluster.

The bond between B and O, r(B-O), is located between the complex with methanol and the complex with the dimethyl-ether. The B-F bond is the longest, compared to covalent complexes with molecules containing oxygen (see Table 3.1 and Table 4.1) but it's the shortest if the complexes with oxides are considered. Similar results in the case of the angle α(OBF) that has the maximum value on molecules containing O and the minimum value when oxides are considered.

Going to investigate the electronic properties of the complex, the situation does not change. The shift of the levels of core 1s of B and F (1.74 and 1.20 eV, respectively) have values closer to those of the molecular complexes (see Table 3.2, Table 3.4 and Table 4.1), as well as the smallest bond energy (0.16 eV).

From the comparison of these parameters, it emerges that the basicity of the SIO_2 is much less pronounced than of the oxides of alkaline-earth metals, and it is placed among the common molecules considered. This is consistent with the value of the ionization potential and the level of the E_{HOMO} (10.33 and 8.31 Ev respectively) which are very close to the values found in molecular complexes (see Table 3.1, Table 3.4 and Table 4.1).

4.3 Correlation between the surface basicity and the adsorbed BF3

We tried to report some properties of the BF_3 deducted from the calculations about the adsorption of this molecule on rutile and quartz, together with the values discussed previously related to molecular and surface complexes.

4.3.1 Geometric properties

The following graphs report some geometric properties including the distance r(B-O), the bond angle α(XBF), the distance r(B-F) versus the ionization potential vertical, Figure 4.6, Figure 4.7, and Figure 4.8, respectively.

It is immediately noticeable that the values of the complexes with TiO_2 and SiO_2, correlate very well with those previously obtained for alkaline-earth metals and for molecular complexes. This is also deduced from the correlation coefficients which improve albeit a little.

In the plot of the bond distance r(B-O) versus IP, the correlation coefficient goes from 0.961 in the plot without the values of the complexes with TiO_2, and SIO_2, Figure 3.7, to 0.963 in the plot with such values included, Figure 4.6.

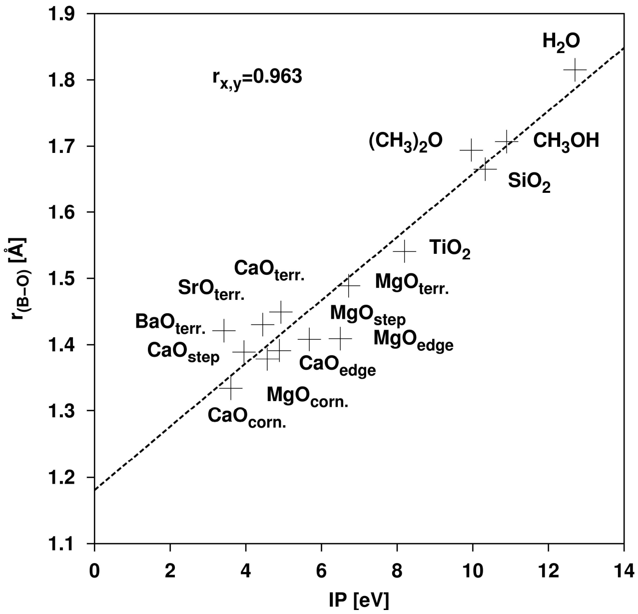

Figure 4.6 – Distance r(B–O) versus to the ionization potential IP.

Also comparing Figure 3.9 and Figure 4.7, which refer to distance r(B-F) versus IP, an improvement of the correlation coefficient, from -0.953 to -0.957, is noted.

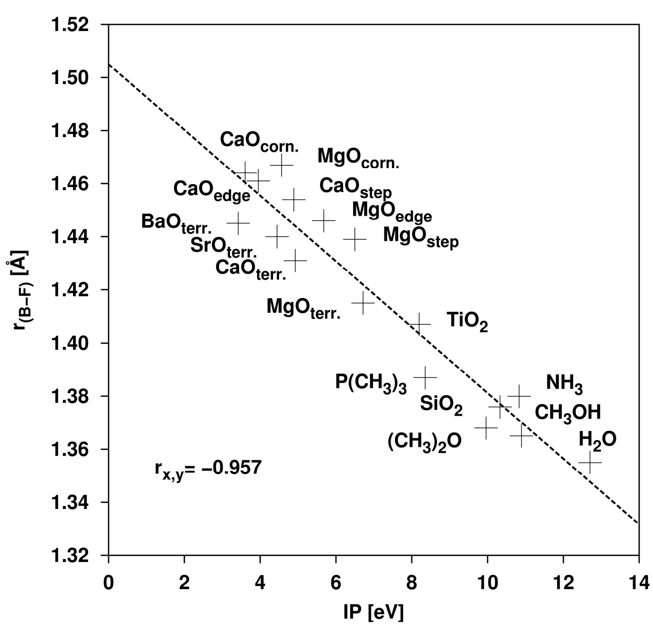

Figure 4.7 – Distance r(B–F) versus to the ionization potential IP.

Only in the comparison between Figure 3.11 and Figure 4.8, referring to the bond angle of α(XBF) vs. IP, the correlation coefficient decreases from -0.974 to -0.972.

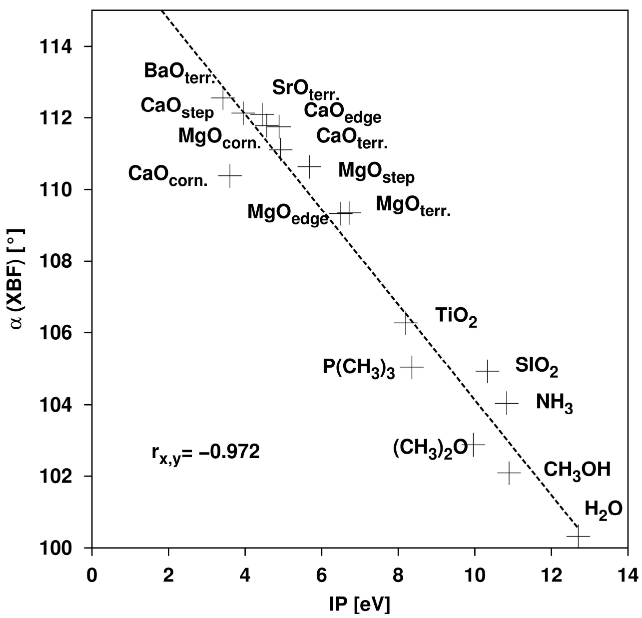

Figure 4.8 – Angle α(XBF) versus to the ionization potential IP.

Finally, it is important to notice where the data of Rutile and silica are placed. The latter is always placed among the molecules, and this is explained by the fact that it is a covalent compound. The rutile, partially ionic, is located halfway between complexes with covalent compounds and those with purely ionic oxides.

4.3.2 Electronic properties

Lastly, the behavior of rutile and silica towards the shifts of the 1s levels of the atoms of B, F, and O versus IP is considered, Figure 4.9, Figure 4.10, and Figure 4.11, respectively. The values regarding silica, as in previous cases, are among those of molecules. Those about the rutile, on the other hand, are completely out of the fitting line since, as previously mentioned, the rutile cluster model is not suitable for treating electronic properties, as an incorrect surface potential is created.

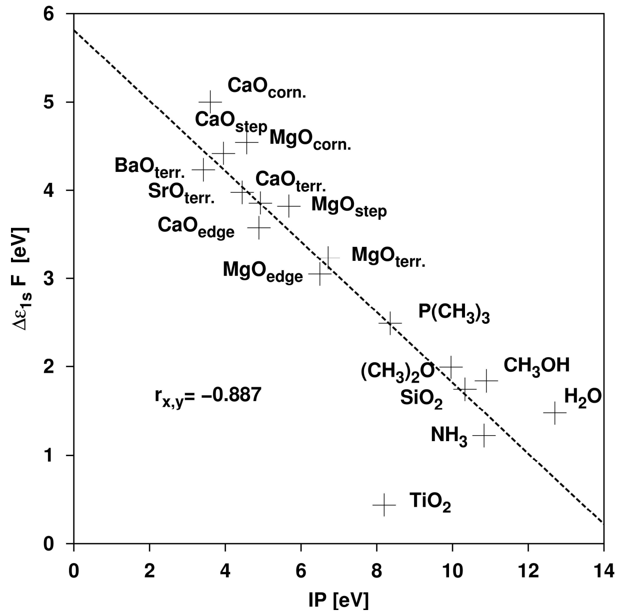

Figure 4.9 – Shift of the F level 1s versus to the ionization potential IP.

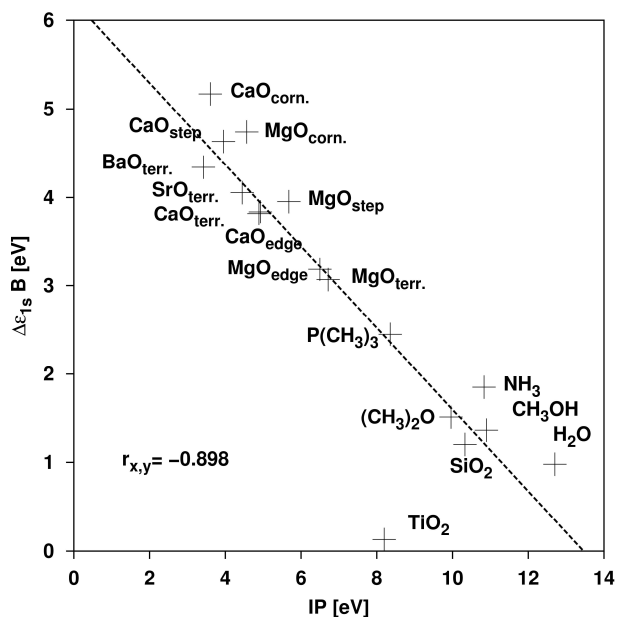

Figure 4.10 – Shift of the B level 1s versus to the ionization potential IP.

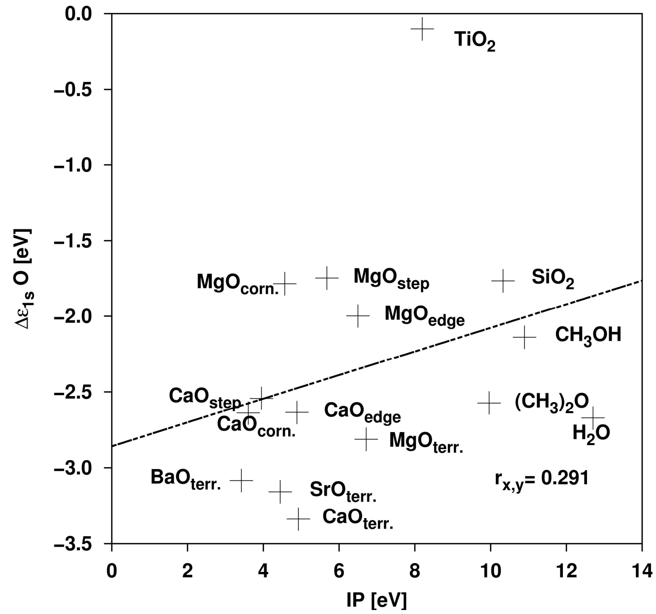

Figure 4.11 – Shift of the oxygen level 1s versus the ionization potential IP.

The only quantity that seems to correlate well also for the rutile is the bond energy D_e. From the plot in Figure 4.12, it is noted that the data point of TiO_2 is close to the fitting line, and it is just above the data related to the molecular complexes. Such an improvement is also shown in the correlation coefficients that go from -0.926 to -0.929 (see Figure 3.21 and Figure 4.12).

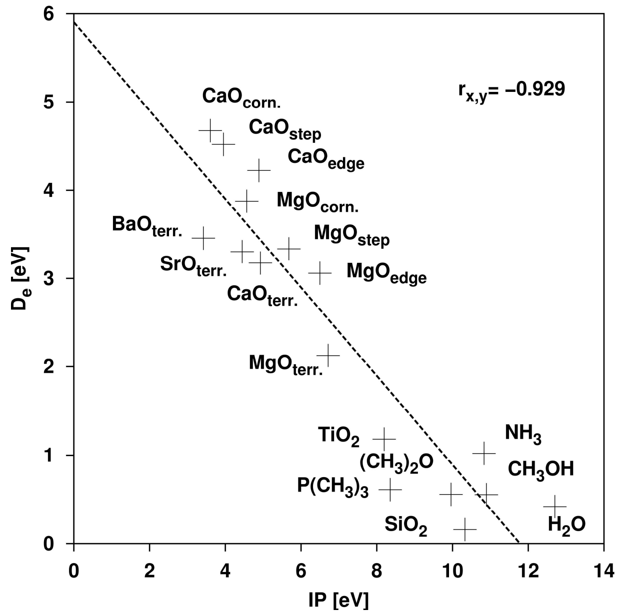

Figure 4.12 – Bond energy D_e versus to the ionization potential IP.

5. Conclusions

The surface basicity of alkaline-earth oxides is a topic of primary interest in catalysis and in surface chemistry. Experimentally it is known that the degree of basicity grows along the MgO < CaO < SrO < BaO series [1]. It has been shown theoretically that this trend is not related to a significant change in character ionic oxides (all these oxides are highly ionic with small deviations by going from MgO to BaO)[37], but rather with a variation in the Madelung potential along the series.[5] The main change from MgO to BaO is the increase of the reticular parameter which in turn depends on the size of the cation (the distance M-O is 2,106 Å in MgO, 2,399 Å in CaO, 2,580 Å in SrO and 2,776 Å in BaO). The increase in the reticular parameter has as consequence a decrease in the Madelung potential of heavier alkaline-earth oxides.

Since the O^{2-} ion is only stable inside the ionic crystals for the effects of the electrostatic field, a decrease in Madelung potential leads to a destabilization of the energy levels of O^{2-} and therefore a greater tendency to donate charge. This simple but convincing explanation applies not only to the various alkaline-earth oxides, but also to the different sites within the same surface oxide. The Madelung potential for a $O^{2-}{}_{6c}$, of bulk of a cubic lattice type MgO is 23.9 eV, 23.0 eV for a terrace $O^{2-}{}_{5c}$, 21.8 eV for a step $O^{2-}{}_{4c}$, and 18.4 eV for a vertex $O^{2-}{}_{3c}$.[38] With respect to this, the decrease in Madelung potential in low coordination sites (edges, steps, vertices, kinks, and so on) results in a higher basic nature of these sites than the oxide anions on flat terraces. This important observation led to the theoretical prediction of a completely different reactivity of probe molecules such as the CO_2 on regular and defective sites of alkaline-earth oxides such as MgO or CaO[5], subsequently confirmed from electronic spectroscopy experiments with metastable ions (MIES)[39,40] of CO_2 adsorption on MgO and CaO, and finally by photoemission measurements with synchrotron light.[41,42] While the CaO surface exposed to the CO_2 is very reactive and forms carbonate complexes, CO_2 chemisorption on MgO takes place only on defective sites, presumably on the O^{2-} ions of steps and vertices. Compared to this, the use of BF_3 as a probe for surface basicity adds nothing new to the known trends, but it is promising to extend the analysis from a more qualitative level (see for example the use of CO_2) to a more quantitative estimate of surface basicity properties. In particular, it was shown that various geometric parameters offer a high level of correlation with the basicity of the adsorption site measured by its IP. Since careful measurements of the distances and bond angles are available, at least In principle, through NEXAFS spectroscopy[43], it is suggested that this measurement can provide a direct way to titrate the donor properties of the system. Another quantity that seems to correlate well with surface basicity is the stretching frequency of the B-F bond: the shifts predicted by the calculations for the various absorption sites considered are large enough to be distinguished in an IR spectroscopy measurement. Less simple is the use of photoemission spectroscopy to identify the level of charge transfer from the surface to the BF_3. About that, it could be said that all the results therein discussed are related to the energy levels of the system before the ionization process took place (initial state effect), and that the conclusions can be different if the screening from final state is included. However, based on the simple analysis of the Kohn-Sham levels, it is found that while the B and F 1s core levels provide a direct measure of the surface basicity, the 1s levels of O show more complex behavior, not easy to be rationalized simply in terms of charge transfer. Unfortunately, the shifts of the 1s levels of B and F compared to the BF_3 in the gaseous phase are not easily accessible. Therefore, it seems that XPS spectroscopy represents a less direct tool to analyze this type of problem. The last quantity considered is the absorption energy of the BF_3 on the oxides surface. It is found that this property shows a sufficiently close correlation to surface basicity, such as to suggest its use for the experimental titration of surfaces. Since the desorption energy in the TDS spectra is related to the strength of bond of an adsorbed species on the surface, it seems that this spectroscopy can be used, in particular to seek the presence of sufficiently high concentrations of low coordination sites.

A general problem that has not been addressed here, but that could limit the use of BF_3 is its tendency to dissociate and to form reactive fragments. With growing the temperature, the possibility that BF_3 dissociates becomes more realistic. In the study on alkaline-earth oxides no spontaneous trend towards dissociation was observed, also on the sites with the most reactive low coordination sites. This is no longer true on other surfaces. Experimentally, it was observed that the bridge oxygens in SnO_2 react with the BF_3.[7] Our calculations on TiO_2 surface (110) confirm this result and show that while the tri-coordinated oxide anions adsorb BF_3, dissociative adsorption takes place on the bridge oxygen with a non-activated process.

Finally, it should be noted that we did not deal with the thermodynamic and kinetic aspects of the dissociative adhesion of the BF_3 on alkaline-earth oxides. Consequently, while the conclusions of this study are valid for low substrate temperatures (< RT), some attention in their extension to greater temperatures is required.

Bibliography

(1) Boudart, M.; Anderson, J. R.; K. Tanabe. *Catalysis and Technology*; springer, 1981.

(2) Zecchina, A.; Scarano, D.; Bordiga, S.; Ricchiardi, G.; Spoto, G.; Geobaldo, F. IR Studies of CO and NO Adsorbed on Well Characterized Oxide Single Microcrystals. *Catal. Today* **1996**, *27* (3), 403–435. https://doi.org/10.1016/0920-5861(95)00202-2.

(3) Pacchioni, G.; Cogliandro, G.; Bagus, P. S. Molecular Orbital Cluster Model Study of Bonding and Vibrations of CO Adsorbed on MgO Surface. *Int. J. Quantum Chem.* **1992**, *42* (5), 1115–1139. https://doi.org/10.1002/qua.560420504.

(4) Nakano, Y.; Iizuka, T.; Hattori, H.; Tanabe, K. Surface Properties of Zirconium Oxide and Its Catalytic Activity for Isomerization of 1-Butene. *J. Catal.* **1979**, *57* (1), 1–10. https://doi.org/10.1016/0021-9517(79)90038-1.

(5) Pacchioni, G.; Ricart, J. M.; Illas, F. Ab Initio Cluster Model Calculations on the Chemisorption of CO2 and SO2 Probe Molecules on MgO and CaO (100) Surfaces. A Theoretical Measure of Oxide Basicity. *J. Am. Chem. Soc.* **1994**, *116* (22), 10152–10158. https://doi.org/10.1021/ja00101a038.

(6) Pacchioni, G. Physisorbed and Chemisorbed CO2 at Surface and Step Sites of the MgO(100) Surface. *Surf. Sci.* **1993**, *281* (1), 207–219. https://doi.org/10.1016/0039-6028(93)90869-L.

(7) Abee, M. W.; Cox, D. F. BF3 Adsorption on α-Cr2O3 $(10\bar{1}2)$: Probing the Lewis Basicity of Surface Oxygen Anions. *J. Phys. Chem. B* **2001**, *105* (35), 8375–8380. https://doi.org/10.1021/jp011389v.

(8) Abee, M. W.; Cox, D. F. BF3 Adsorption on Stoichiometric and Oxygen-Deficient SnO2(110) Surfaces. *J. Phys. Chem. B* **2003**, *107* (8), 1814–1820. https://doi.org/10.1021/jp021771n.

(9) Maria, P. C.; Gal, J. F. A Lewis Basicity Scale for Nonprotogenic Solvents: Enthalpies of Complex Formation with Boron Trifluoride in Dichloromethane. *J. Phys. Chem.* **1985**, *89* (7), 1296–1304. https://doi.org/10.1021/j100253a048.

(10) Rauk, A.; Hunt, I. R.; Keay, B. A. Lewis Acidity and Basicity: An Ab Initio Study of Proton and BF3 Affinities of Oxygen-Containing Organic Compounds. *J. Org. Chem.* **1994**, *59* (22), 6808–6816. https://doi.org/10.1021/jo00101a049.

(11) Becke, A. D. Density-Functional Exchange-Energy Approximation with Correct Asymptotic Behavior. *Phys. Rev. A* **1988**, *38* (6), 3098–3100. https://doi.org/10.1103/PhysRevA.38.3098.

(12) Becke, A. D. Density-functional Thermochemistry. III. The Role of Exact Exchange. *J. Chem. Phys.* **1993**, *98* (7), 5648–5652. https://doi.org/10.1063/1.464913.

(13) Lee, C.; Yang, W.; Parr, R. G. Development of the Colle-Salvetti Correlation-Energy Formula into a Functional of the Electron Density. *Phys. Rev. B Condens. Matter* **1988**, *37* (2), 785–789. https://doi.org/10.1103/physrevb.37.785.

(14) Ahlrichs, R.; Taylor, P. R. The Choice of Gaussian Basis Sets for Molecular Electronic Structure Calculations. *J. Chim. Phys.* **1981**, *78*, 315–324. https://doi.org/10.1051/jcp/1981780315.

(15) Raffenetti, R. C. General Contraction of Gaussian Atomic Orbitals: Core, Valence, Polarization, and Diffuse Basis Sets; Molecular Integral Evaluation. *J. Chem. Phys.* **1973**, *58* (10), 4452–4458. https://doi.org/10.1063/1.1679007.

(16) Hay, P. J.; Wadt, W. R. Ab Initio Effective Core Potentials for Molecular Calculations. Potentials for the Transition Metal Atoms Sc to Hg. *J. Chem. Phys.* **1985**, *82* (1), 270–283. https://doi.org/10.1063/1.448799.

(17) Wadt, W. R.; Hay, P. J. Ab Initio Effective Core Potentials for Molecular Calculations. Potentials for Main Group Elements Na to Bi. *J. Chem. Phys.* **1985**, *82* (1), 284–298. https://doi.org/10.1063/1.448800.

(18) Hay, P. J.; Wadt, W. R. Ab Initio Effective Core Potentials for Molecular Calculations. Potentials for K to Au Including the Outermost Core Orbitals. *J. Chem. Phys.* **1985**, *82* (1), 299–310. https://doi.org/10.1063/1.448975.

(19) Hehre, W. J.; Ditchfield, R.; Pople, J. A. Self—Consistent Molecular Orbital Methods. XII. Further Extensions of Gaussian—Type Basis Sets for Use in Molecular Orbital Studies of Organic Molecules. *J. Chem. Phys.* **1972**, *56* (5), 2257–2261. https://doi.org/10.1063/1.1677527.

(20) Maria Ferrari, A.; Soave, R.; D'Ercole, A.; Pisani, C.; Giamello, E.; Pacchioni, G. Theoretical Characterization of Charge-Transfer Reactions between N2 and O2 Molecules and Paramagnetic Oxygen Vacancies on the MgO Surface. *Surf. Sci.* **2001**, *479* (1), 83–97. https://doi.org/10.1016/S0039-6028(01)00946-3.

(21) Stevens, W. J.; Basch, H.; Krauss, M. Compact Effective Potentials and Efficient Shared-exponent Basis Sets for the First- and Second-row Atoms. *J. Chem. Phys.* **1984**, *81* (12), 6026–6033. https://doi.org/10.1063/1.447604.

(22) Frisch, M. J. Gaussian 98, 1998.

(23) Lide, D. R.; Data, S. R.; Board, E. A.; Baysinger, G.; Chemistry, S.; Library, C. E.; Berger, L. I.; Goldberg, R. N.; Division, B.; Kehiaian, H. V.; Kuchitsu, K.; Rosenblatt, G.; Roth, D. L.; Zwillinger, D. *CRC Handbook of Chemistry and Physics*. https://doi.org/978-1466571143.

(24) Herzberg, G. *Molecular Spectra and Molecular Structure. II. Infrared and Raman Spectra of Polyatomic Molecules*; Van Nostrand Reinhold: Nre York, 1945.

(25) Bagus, P. S.; Brundle, C. R.; Pacchioni, G.; Parmigiani, F. Mechanisms Responsible for the Shifts of Core-Level Binding Energies between Surface and Bulk Atoms of Metals. *Surf. Sci. Rep.* **1993**, *19* (3), 265–283. https://doi.org/10.1016/0167-5729(93)90013-F.

(26) Brown, H. C. Chemical Effects of Steric Strains. *J. Chem. Soc. Resumed* **1956**, No. 0, 1248–1268. https://doi.org/10.1039/JR9560001248.

(27) Ahlrichs, R.; Bär, M. R.; Häser, M.; Sattler, E. Ab Initio Investigation of the Structures and Stabilities of the Donor-Acceptor Complexes H3PBF3 and Me3PBF3. *Chem. Phys. Lett.* **1991**, *184* (4), 353–358. https://doi.org/10.1016/0009-2614(91)85136-K.

(28) Barr, T. L.; Brundle, C. R. Bonding and Electronic Structure in High-${\mathit{T}}_{\mathit{c}}$ Superconducting Oxides: A Case for the Importance of the Cations. *Phys. Rev. B* **1992**, *46* (14), 9199–9204. https://doi.org/10.1103/PhysRevB.46.9199.

(29) Pacchioni, G.; Bagus, P. S. Theoretical Analysis of the O(1s) Binding-Energy Shifts in Alkaline-Earth Oxides: Chemical or Electrostatic Contributions. *Phys. Rev. B Condens. Matter* **1994**, *50* (4), 2576–2581. https://doi.org/10.1103/physrevb.50.2576.

(30) Cox, P. A. *Transition Metal Oxides: An Introduction to Their Electronic Structure and Properties*; The International Series of Monographs on Chemistry; Clarendon Press: Oxford, 1995; Vol. 27.

(31) Sauer, J. Molecular Models in Ab Initio Studies of Solids and Surfaces: From Ionic Crystals and Semiconductors to Catalysts. *Chem. Rev.* **1989**, *89* (1), 199–255. https://doi.org/10.1021/cr00091a006.

(32) Bredow, T. Effect of Embedding and Cluster Size on the Ab Initio Study of Potassium Adsorption at Rutile(110). *Int. J. Quantum Chem.* **1999**, *75* (2), 127–132. https://doi.org/10.1002/(SICI)1097-461X(1999)75:2<127::AID-QUA6>3.0.CO;2-R.

(33) Bredow, T.; Aprà, E.; Catti, M.; Pacchioni, G. Cluster and Periodic Ab-Initio Calculations on K/TiO2(110). *Surf. Sci.* **1998**, *418* (1), 150–165. https://doi.org/10.1016/S0039-6028(98)00712-2.

(34) Heise, R.; Courths, R.; Witzel, S. Valence Band Densities-of-States of TiO2(110) from Resonant Photoemission and Photoelectron Diffraction. *Solid State Commun.* **1992**, *84* (6), 599–602. https://doi.org/10.1016/0038-1098(92)90198-I.

(35) Pacchioni, G.; Ieranò, G. Computed Optical Absorption and Photoluminescence Spectra of Neutral Oxygen Vacancies in $\mathit{\ensuremath{\alpha}}$-Quartz. *Phys. Rev. Lett.* **1997**, *79* (4), 753–756. https://doi.org/10.1103/PhysRevLett.79.753.

(36) Pacchioni, G.; Ieranò, G. Ab Initio Formation Energies of Point Defects in Pure and Ge-Doped ${\mathrm{SiO}}_{2}$. *Phys. Rev. B* **1997**, *56* (12), 7304–7312. https://doi.org/10.1103/PhysRevB.56.7304.

(37) Pacchioni, G.; Sousa, C.; Illas, F.; Parmigiani, F.; Bagus, P. S. Measures of Ionicity of Alkaline-Earth Oxides from the Analysis of Ab Initio Cluster Wave Functions. *Phys. Rev. B* **1993**, *48* (16), 11573–11582. https://doi.org/10.1103/PhysRevB.48.11573.

(38) Sauer, J.; Ugliengo, P.; Garrone, E.; Saunders, V. R. Theoretical Study of van Der Waals Complexes at Surface Sites in Comparison with the Experiment. *Chem. Rev.* **1994**, *94* (7), 2095–2160. https://doi.org/10.1021/cr00031a014.

(39) Ochs, D.; Brause, M.; Braun, B.; Maus-Friedrichs, W.; Kempter, V. CO2 Chemisorption at Mg and MgO Surfaces: A Study with MIES and UPS (He I). *Surf. Sci.* **1998**, *397* (1), 101–107. https://doi.org/10.1016/S0039-6028(97)00722-X.

(40) Ochs, D.; Braun, B.; Maus-Friedrichs, W.; Kempter, V. CO2 Chemisorption at Ca and CaO Surfaces: A Study with MIES, UPS(HeI) and XPS. *Surf. Sci.* **1998**, *417* (2), 406–414. https://doi.org/10.1016/S0039-6028(98)00721-3.

(41) Carrier, X.; Doyle, C. S.; Kendelewicz, T.; Brown, G. E. REACTION OF CO2 WITH MgO(100) SURFACES. *Surf. Rev. Lett.* **1999**, *06* (06), 1237–1245. https://doi.org/10.1142/S0218625X99001396.

(42) Doyle, C. S.; Kendelewicz, T.; Carrier, X.; Brown, G. E. THE INTERACTION OF CARBON DIOXIDE WITH SINGLE CRYSTAL CaO(100) SURFACES. *Surf. Rev. Lett.* **1999**, *06* (06), 1247–1254. https://doi.org/10.1142/S0218625X99001402.

(43) Hoeft, J.-T.; Kittel, M.; Polcik, M.; Bao, S.; Toomes, R. L.; Kang, J.-H.; Woodruff, D. P.; Pascal, M.; Lamont, C. L. A. Molecular Adsorption Bond Lengths at Metal Oxide Surfaces: Failure of Current Theoretical Methods. *Phys. Rev. Lett.* **2001**, *87* (8), 086101. https://doi.org/10.1103/PhysRevLett.87.086101.